Protection and Dilution: SAFEs and MFN Clauses in Seed Financing

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Abstract

We ask how an accelerator should protect its stake against the seed investor who finances its startups next, and what a most-favored-nation (MFN) clause, such as the $375,000 uncapped MFN SAFE that Y Combinator added to its standard deal in 2022, is worth to it. In our model, the accelerator spends its contribution before a liquidity shock, and a seed investor with market power covers the rest of the shock with a priced or an unpriced SAFE. Without protection, the seed investor takes all of the equity of the projects it finances, except with an uncapped SAFE without discount, which it never uses: the accelerator keeps value only on the shocks it covers alone. A fixed post-money share gives the accelerator this share of every project the seed investor finances. Any fixed share that comes with a contribution is strictly beaten by an MFN clause with a small MFN component, whether the seed investor uses a priced or an unpriced SAFE: the component takes more from small rounds than from large ones. Against the clause, the seed investor prefers the unpriced SAFE, and answers by investing beyond its need in the smallest rounds. Whatever the protection, a seed round covers at most the project's expected surplus at the scaling cost, which is below the scaling cost. Numerically, the accelerator invests, and puts almost all of its money through the clause, only when liquidity needs are small relative to the scaling cost. For larger needs, it buys a fixed share with an amount and at a valuation that tend to zero, and this share lowers the total surplus below the first best, which the absence of protection reaches.

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1 Introduction

Most accelerators offer the same standard deal to every startup in a batch. The most common deal is a small amount for a share of equity, which Techstars offered from its first program in 2007.[1] Y Combinator (YC) put on average about $17,000 in equity for 7% and $80,000 in a safe converting at the terms of the next money raised.[2] In 2014, it replaced this structure by a single deal, "$120k for 7% equity (regardless of the number of founders)", to avoid complexity.[3] Sam Altman, president of YC in 2014, argued: "Speaking of hating complexity, we've tried to make the new structure really simple. The convertible notes and safes we used got complicated in terms of how they got priced, and complexity often causes unintended consequences. It was hard for founders to actually predict how much total dilution they were looking at." But, in January 2022, YC added to its $125,000 for 7% a $375,000 uncapped Simple Agreement for Future Equity (SAFE) with a most-favored-nation (MFN) clause: the SAFE converts at the best terms the startup grants to intermediate investors before its first equity round. The safe of the deal before 2014 converted at the terms of the next money raised, that is, at the terms of the first of these investors only, so that the two deals coincide when a single seed round precedes the equity round, as in our model, and differ when several seed rounds follow. The deal is still the same for every startup in the batch, and according to Geoff Ralston, president of YC in 2022, it should "remove the immediate pressure to fundraise and accept less than favorable terms".[4] Three quarters of YC's money now go through the clause.

The MFN clause answers a structural problem of sequential venture financing. The accelerator sources and trains startups and hands them to venture capitalists (VCs) (Gonzalez-Uribe and Leatherbee, 2018; Hallen et al., 2020). When it invests, neither the startup's probability of success nor its exact financing need is known, an uncertainty that comes with early-stage financing (Nanda and Rhodes-Kropf, 2017; Lerner and Nanda, 2020; Ewens et al., 2018). Information arrives in sequence: the financing need is known before the quality (Kerr et al., 2014; Ewens et al., 2018; Bergemann and Hege, 1998, 2005). A startup from an accelerator may therefore need additional funding before it can raise an equity round (Coyle and Green, 2014), and the intermediate investors who provide it have market power over the entrepreneur. The terms they set dilute the accelerator, and a protection written before the need is known decides how much of the surplus the accelerator keeps. Nondiscrimination clauses such as MFN clauses are meant to protect early signatories against this opportunism, but McAfee and Schwartz (1994) show that they generally fail to do so. Ralston also announced

[1] "Initially, Techstars invested between $6,000 and $18,000 in early stage companies" (Techstars, Wikipedia).

[2] TechCrunch, April 22, 2014.

[3] Sam Altman, "The New Deal," Y Combinator blog, April 22, 2014, which is also the source of the quotation below.

[4] "YC's $500,000 Standard Deal," Y Combinator blog, January 11, 2022, `https://www.ycombinator.com/blog/ycs-500-000-standard-deal`.

that the new deal should help founders "raise a successful seed round".[5] This paper asks which protection an accelerator should write against the seed investor, within the family of SAFEs that accelerators and seed investors use, what an MFN clause is worth to it, and how the clause changes the total surplus, which we use to measure welfare.

We summarize our findings. First, without protection, the seed investor takes all of the equity of the projects it finances. The accelerator's contribution is spent before the shock, and the accelerator keeps value only on the shocks it covers alone (Propositions 1 and 3). The seed investor obtains the same with a capped SAFE or with an uncapped SAFE whose discount it chooses. An uncapped SAFE without discount pays back at most its amount, and the seed investor never uses it (Proposition 2). Second, a fixed post-money share gives the accelerator this share of every project the seed investor finances, and the accelerator earns strictly more than without protection (Propositions 4–6). Priced and unpriced SAFEs give the same outcome (Proposition 5). A capped SAFE for the accelerator does no better than the fixed share. It does as well when the fixed share keeps more than what the SAFE returns of its amount, since the seed investor invests only its need (Proposition 7). Third, every fixed share that comes with a contribution is strictly beaten by an MFN clause with the same contribution and a small MFN component, whether the seed investor uses an unpriced SAFE (Proposition 9) or a priced one (Proposition 18). When the optimal fixed share comes with a contribution, the optimal protection therefore includes an MFN component. The clause takes a share that falls with the round: on a small round, the accelerator's money weighs a lot against the seed investor's. Against the clause, the seed investor prefers the unpriced SAFE, which converts at the same price as the MFN component in every state (Proposition 8). It answers by investing beyond its need in the smallest rounds, and every dollar it adds is lost on the projects that are not fully funded. The fixed part of the clause curbs this answer, and the clause is bounded by the accelerator's contribution: in our simulations, almost all of the contribution goes through the clause, where more protection would still pay. Fourth, a seed round covers at most the project's expected surplus at the scaling cost, which is itself below the scaling cost (Proposition 10). A protection leaves the accelerator a share of the last project the seed investor finances, and the seed investor stops earlier (Proposition 11). A protection changes the probabilities of a seed round and of full funding through the contribution, which moves the seed investor's zone, and through what the accelerator keeps on the last project, which shortens it; the seed investor's investment beyond its need raises the probability of full funding (Propositions 12 and 13). The total surplus is maximal when the accelerator contributes nothing, and without protection it falls with the contribution. The protections move it either way, and the investment beyond the need lowers it (Propositions 15–17). Fifth, numerically, the two regimes separate sharply (Section 8). When liquidity needs are small relative to the scaling cost, $E(\ell)/I \leq 0.05$, the accelerator invests, almost all of its money goes through the clause, and the MFN clause gives the highest profit

[5]Same source. The MFN SAFE "will take on the terms of the lowest cap safe (or other most favorable terms)" issued "between the start of the batch and the next equity round".

and the highest total surplus. Whether the fixed part of the clause is a share or the accelerator's own SAFE makes no difference (Section 9). From $E(\ell)/I = 0.15$ on, the accelerator's optimal contribution vanishes under any protection: it buys a fixed share with an amount and at a valuation that tend to zero, and this share lowers the total surplus to 0.74–0.77 of the first best, which the absence of protection reaches.

To arrive at these conclusions, we set up a three-date game between risk-neutral agents. We see entrepreneurship as experimentation (Kerr et al., 2014): a project must survive a liquidity shock before its value is revealed, and is worth funding only once the value is known. The entrepreneur sells the whole project to the accelerator, which spends a fixed amount with a protection before the shock. The seed investor observes the shock and makes a take-it-or-leave-it offer to cover it, with a priced or an unpriced SAFE. Once the value is revealed, a competitive market funds the projects worth funding through equity rounds. Money invested in the project never returns to its investors, and a rejected seed round abandons the project: the accelerator accepts any round. We solve the game by backward induction, first without protection (Section 3), then with a fixed post-money share or a capped SAFE for the accelerator (Section 4), then with an MFN component (Section 5). We then derive the probabilities of a seed round and of full funding (Section 6) and the total surplus (Section 7). All the analytical results hold for any log-concave distributions of the project's value and of the liquidity shock. Section 8 computes the optimal contracts for software and deep-tech projects, and Section 9 checks them against another distribution of the project's value and a stake kept by the founders, and extends them to a seed investor that uses a priced SAFE against the MFN clause and to the accelerator's own SAFE as the fixed part of the clause. Section 10 discusses the seed investor's market power, the accelerator's contribution and the contracts we leave aside.

Our contribution is twofold. First, we propose a tractable model of the accelerator's protection against a seed investor with market power, in which the protection is a cap on the share the seed investor may take, possibly depending on the amount it invests. Fixed shares and MFN clauses are special cases. To our knowledge, it is the first model of an accelerator's protection against seed dilution. Second, we evaluate the MFN clause that accelerators adopted after YC. A small MFN component improves on any fixed share that comes with a contribution, because it discriminates between rounds, and its value is limited by the seed investor's answer and by the accelerator's own contribution, which caps the amount the clause covers. Its effect on the total surplus is not signed in general: numerically, the clause raises it for software-like projects, and any protection lowers it for deep-tech projects.

Our paper relates to four literatures.

The first studies entrepreneurship as experimentation. Kerr et al. (2014) argue that the probability of success of a new venture is unknown until someone invests in it, and that the cost of experimenting shapes who finances innovation. Ewens et al. (2018) show that when cloud computing lowered this cost, VCs started to give little money and little governance to many more startups, and to abandon more of them. Nanda

and Rhodes-Kropf (2017) show that investors can protect a startup against a lack of future funding by giving it more money upfront, at the cost of the option to stop. Investors also stage their financing to learn before investing more. Gompers (1995) shows that VCs stage more often when agency costs are higher. Neher (1999) explains staging by the entrepreneur's ability to renegotiate once the money is invested, early rounds building the collateral that supports later ones. Bergemann and Hege (1998, 2005) show that funding stops too early when the learning process is subject to moral hazard. We add to this literature by modeling the terms on which the experiment's cost is financed after the first investor has invested, and how the first investor protects itself.

The second studies the securities used in venture capital. Cornelli and Yosha (2003) show that convertible debt deters the entrepreneur from manipulating interim results to avoid liquidation, because a better interim result also makes the VC convert. Schmidt (2003) shows that convertible securities give both the entrepreneur and the VC incentives to invest in the project. Repullo and Suarez (2004) derive the optimal securities when the project is financed in two stages under double-sided moral hazard. In Casamatta (2003), the VC both finances and advises the firm, and receives common stock when it invests little and convertible securities when it invests a lot. Hellmann (1998) explains why entrepreneurs give VCs the right to replace them. Admati and Pfleiderer (1994) show that a VC who receives a fixed fraction of the payoff and finances the same fraction of every later round has no incentive to misprice later rounds. Kaplan and Strömberg (2003, 2004) document that VC contracts allocate cash-flow, board, voting and liquidation rights separately, and that riskier deals give the VC more of these rights. Ewens et al. (2022) estimate that VCs use their bargaining power to obtain terms more favorable to investors than the contract that maximizes the startup's success. Coyle and Green (2014) document that after 2005, convertible notes, until then used for bridge loans, became a seed financing instrument with a conversion price cap, followed by the SAFE. Green and Coyle (2016) argue that the SAFE suits only the startups that will raise an equity round, since it converts only then. We add to this literature by comparing a priced and an unpriced SAFE when an accelerator has invested before the seed investor. The two contracts give the same outcome against a fixed share, which caps what an unpriced SAFE takes when the next equity round values the project below its cap. Against an MFN clause, the seed investor prefers the unpriced SAFE, which converts at the same price as the accelerator's MFN component.

The third studies most-favored-customer clauses. Cooper (1986) shows that a firm that promises to rebate its past customers if it later cuts its price has less reason to cut it, so the clause helps duopolists keep prices high. Besanko and Lyon (1993) show that a contemporaneous clause, which only covers customers served at the same time, is a weaker device, and that not adopting it can be a dominant strategy. McAfee and Schwartz (1994) show that when a supplier contracts with competing buyers, nondiscrimination clauses generally do not protect a buyer against the supplier's renegotiation with the others, which can explain why franchise contracts are uniform.

In these papers the clause protects a buyer. We add to this literature by studying a clause held by an investor against a later investor. Like the clause of Cooper (1986), YC's MFN SAFE gives an early signatory the better terms granted later. It is a price discrimination device: it takes more from the later investor when that investor's round is small.

The fourth studies the investors who finance a startup before VCs. Hellmann and Thiele (2015) model startups that raise angel money before VC money. The two investors are friends because each relies on the other's investment, and foes because VCs no longer need the angels at the later stage. Hellmann and Puri (2002) show that VC-backed startups professionalize faster, and Hochberg et al. (2007) show that the startups of better-networked VCs are more likely to raise a next round. Cohen and Hochberg (2014) define accelerators as fixed-term programs, of about three months, that support cohorts of startups through the creation of the venture. On their effects, Yu (2020) finds that accelerated startups close down earlier and more often, and raise less money when they close. Gonzalez-Uribe and Leatherbee (2018) find with Start-Up Chile that entrepreneurship schooling improves venture performance, while funding and coworking space alone do not. Hallen et al. (2020) find that some, but not all, accelerators speed up venture development. Kim and Wagman (2014) model an accelerator that chooses its portfolio size and which of its signals to disclose, and Zarei et al. (2022) model the accelerator's selection, mentorship and investment decisions. These papers do not study the terms of the accelerator's investment. We add to this literature by modeling the accelerator's contract and its protection against the seed investor who follows.

2 Model

A project has two sunk costs: a liquidity shock $\ell \geq 0$, with density g and distribution function G, unknown ex ante; and a scaling cost $I > 0$, known ex ante. Its valuation $\pi \in [0, 1]$, with density f and distribution function F, is unknown ex ante. We write

$$W(x) \equiv \int_x^1 (\pi - x)\, f(\pi)\, d\pi,$$

with $W'(x) = -(1 - F(x))$, and $W(x) = E(\pi) - x$ for $x \leq 0$. We make four assumptions:

(A1) $E(\pi) < I$.
(A2) f and g are positive and log-concave.
(A3) When all information is available, all NPV-positive projects obtain funding from a competitive market through equity rounds.
(A4) If ℓ is not covered, π is not revealed.

By (A1), no one funds a project before its value is known: the project must first be an experiment. Since $(\pi - I)^+ \leq \pi$, $W(I) \leq E(\pi)$, and (A1) also gives $W(I) < I$.

Two investors can finance the experiment. The entrepreneur initially sells all of the project to the accelerator, which invests σ_a on each project before ℓ and π are known, with a protection described below. The project spends this money: it never returns to the accelerator. If $\ell > \sigma_a$, a seed investor invests σ_s after ℓ and before π, in exchange for $\rho_s N_t$ shares, where N_t is the number of shares available at date t, and we assume that it has the bargaining power:

(B1) The seed investor proposes take-it-or-leave-it offers.

An equity round that occurs at date t sets a price p_t per share and a number N_t of shares to transfer an amount σ in exchange for ρN_t shares. If an equity round has already occurred, $N_t > N_{t-1}$. We denote by $V_t = N_t p_t = \sigma/\rho$ the post-money valuation, and by $N_{t-1}p_t$ the pre-money valuation. If the accelerator's round is an equity round, N_0 is set at $t = 0$; otherwise, N_0 is set by the first equity round.

Once the seed stage is over, let $x = I + \ell - \sigma_a - \sigma_s$ be the threshold valuation above which the project is fully funded. By (A3), once π is revealed, a competitive investor funds x for the share x/π when $\pi > x$, in an equity round with pre-money valuation $\pi - x$, and the project is abandoned otherwise. An abandoned project returns nothing: the accelerator and the seed investor lose all the money they invested, including the cash beyond the need. The earlier shareholders hold equity worth $W(x)$ in expectation. If $\ell \leq \sigma_a$, the accelerator's cash covers the shock and the threshold is $I + \ell - \sigma_a$.

Before the equity round, the investors use SAFEs. The seed investor offers either a priced SAFE, which fixes its share ρ_s when it invests,[6] or an unpriced SAFE with cap $V_s \geq \sigma_s$ and discount $\lambda \in (0, 1]$, which converts at this equity round at the lower of V_s and $\lambda(\pi - x)$: it claims the share $\sigma_s / \min(V_s, \lambda(\pi - x))$ of the equity held before the round. The accelerator protects its stake by a cap on the share the seed investor may take. We study three protections:

- a fixed post-money share ρ_a: a priced post-money SAFE that buys ρ_a with σ_a, at the post-money valuation σ_a/ρ_a, so that the seed investor's share may not exceed $1 - \rho_a$;
- a capped SAFE for the accelerator: its contribution claims $\sigma_a / \min(V_a, \lambda_a(\pi - x))$ of the equity, with cap $V_a \geq \sigma_a$ and discount $\lambda_a \in (0, 1]$;
- an MFN clause: the accelerator keeps at least ρ_a^0, with a priced post-money SAFE on the rest of its contribution, $\sigma_a - m$, and an amount $m \leq \sigma_a$ converts on the seed investor's terms, at the lower of its valuation and $\pi - x$ for a priced SAFE, and with its cap and discount for an unpriced SAFE.

The accelerator keeps the residual share. Its protection comes first, as in Y Combinator's post-money SAFE, under which "the safes are not diluted by each other":[7]

[6] In this framework, an equity round that sells the share ρ_s is equivalent to a priced SAFE.

[7] Y Combinator, Post-Money Safe User Guide. The dilution from later SAFEs falls on the founders, and the standard SAFE does not say what happens when they have no equity left. The only pro rata rule of the SAFE applies when the proceeds of a liquidity or dissolution event fall

the SAFEs signed after it share the equity it leaves. When they claim more than this equity, each keeps the same fraction of its claim.

The accelerator owns the project and sets the valuation of its priced SAFE, so the share it keeps does not depend on the amount that buys it. A SAFE with no amount buys nothing: a fixed share with $\sigma_a = 0$, or a fixed part with $m = \sigma_a$, is the limit of priced SAFEs whose amount and valuation tend to zero. The accelerator's profit is continuous in these amounts, so it comes as close as it wishes to the profit of the limit, which we treat as a contract.

If the accelerator rejects the seed round, the project is abandoned, and the accelerator, whose contribution is spent, receives nothing. It therefore accepts any round. The seed investor invests only if it earns a nonnegative profit. The timing is as follows. At $t = 0$, the accelerator invests σ_a with its protection. The shock ℓ is then realized. At $t = 1$, the seed investor offers (ρ_s, σ_s), and the accelerator accepts or rejects. The value π is then revealed, and at $t = 2$ the project is fully funded if $\pi > x$.

3 Benchmark: no protection

Proposition 1. *Without protection, for $\ell > \sigma_a$, the seed investor invests if and only if $\ell - \sigma_a \leq W(I)$. It then invests $\sigma_s^* = \ell - \sigma_a$, takes $\rho_s^* = 1$, and earns $W(I) - (\ell - \sigma_a)$; the accelerator keeps nothing.*

→ Proof in Appendix A, page 32

The accelerator's contribution, already spent, reduces the seed investor's need: the seed investor finances shocks up to $\sigma_a + W(I)$, and takes the whole project.

The seed investor may also use an unpriced SAFE. Without protection, the SAFE takes its claim as long as it does not exceed the whole equity: with cap V_s and discount λ, it receives $\min\big(\sigma_s / \min(V_s, \lambda(\pi - x)),\, 1\big)(\pi - x)$ in state $\pi > x$.

Proposition 2. *Without protection, for $\ell > \sigma_a$:*

- *(i) with an uncapped SAFE without discount, the seed investor loses $\int_0^{\sigma_s} F(x + t)\, dt > 0$ at every amount $\sigma_s > 0$, and finances no shock;*
- *(ii) with an uncapped SAFE whose discount it chooses, or with a SAFE whose cap and discount it chooses, it obtains what it obtains with a priced SAFE (Proposition 1), on the same projects.*

In both cases, the accelerator keeps nothing on the shocks $\ell > \sigma_a$.

→ Proof in Appendix A, page 32

short of the amounts due.

An uncapped SAFE without discount receives $\min(\sigma_s, \pi - x)$: it is a debt of face value σ_s, which pays back at most its amount. A discount raises the face value to σ_s/λ, and a low enough discount, or a cap below the amount, gives the seed investor the whole equity.

The accelerator therefore keeps value only on the projects it finances alone, and solves

$$\max_{\sigma_a \geq 0} B(\sigma_a) = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

Proposition 3. *Every optimal contribution σ_a^B is zero or satisfies*

$$W(I)\, g(\sigma_a^B) = \big(1 - G(\sigma_a^B)\big) + \int_0^{\sigma_a^B} F(I + \ell - \sigma_a^B)\, g(\ell)\, d\ell.$$

→ Proof in Appendix A, page 33

The left side is the gain on the project $\ell = \sigma_a$, which moves from the seed investor, who takes all of it, to the accelerator. The right side is the loss of the extra dollar: on the shocks above the contribution, it goes to the seed investor or is lost on an abandoned project, and on the projects the accelerator finances alone, it is returned only when the project is fully funded.

4 A fixed post-money share

4.1 The seed investor

Proposition 4 (Priced SAFE)**.** *With a fixed share $\rho_a > 0$, for $\ell > \sigma_a$, the seed investor invests $\sigma_s^* = \ell - \sigma_a$, the threshold is I, and it takes $1 - \rho_a$. It earns $(1 - \rho_a)W(I) - (\ell - \sigma_a)$, and invests if and only if $\ell \leq \sigma_a + (1 - \rho_a)W(I)$; the accelerator keeps $\rho_a W(I)$.*

→ Proof in Appendix A, page 33

Proposition 5 (Unpriced SAFE)**.** *For $\ell > \sigma_a$, the seed investor obtains with an unpriced SAFE exactly what it obtains with a priced SAFE, on the same projects, and the discount plays no role. The cap $V_s^* = (\ell - \sigma_a)/(1 - \rho_a)$ gives it the share $1 - \rho_a$ in every state above I.*

→ Proof in Appendix A, page 33

An unpriced SAFE converts at the lower of its cap and the discounted pre-money valuation of the next equity round: when this round values the project below the cap, the SAFE returns at least its amount. The fixed share caps what the seed investor takes in these states, and a cap low enough gives it the share $1 - \rho_a$ in every state, as a priced SAFE does.

4.2 The accelerator

With a fixed share, the accelerator's profit is

$$\Pi_a(\sigma_a, \rho_a) = B(\sigma_a) + \rho_a W(I)\Big[G\big(\sigma_a + (1-\rho_a)W(I)\big) - G(\sigma_a)\Big]$$

against a priced or an unpriced SAFE.

Proposition 6. *(i) At each σ_a, the optimal share $\rho_a^*(\sigma_a)$ is unique, lies in $(0,1)$, and satisfies*

$$G\big(\sigma_a + (1-\rho_a)W(I)\big) - G(\sigma_a) = \rho_a W(I)\, g\big(\sigma_a + (1-\rho_a)W(I)\big).$$

(ii) Every optimal contribution σ_a^* is zero or satisfies

$$(1-\rho_a^*)W(I)\, g(\sigma_a^*) + \rho_a^* W(I)\, g\big(\sigma_a^* + (1-\rho_a^*)W(I)\big) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

(iii) The accelerator earns strictly more than in the benchmark.

→ Proof in Appendix A, page 34

A dollar more moves the whole zone of the seed investor by one dollar. The project $\ell = \sigma_a$ moves from the seed investor to the accelerator, which gains the seed investor's share of it, $(1-\rho_a)W(I)$, and a project enters at $\sigma_a + (1-\rho_a)W(I)$, where the accelerator keeps $\rho_a W(I)$. At a given contribution, the marginal value of the contribution differs from the benchmark by $\rho_a W(I)[g(\sigma_a + (1-\rho_a)W(I)) - g(\sigma_a)]$: the fixed share pushes the contribution up when the shock density is higher at the last project than at the first, and down otherwise.

4.3 A capped SAFE for the accelerator

From 2019, Y Combinator's \$150,000 for 7% was such a SAFE, a post-money SAFE whose cap gave it at least 7% (Section 9). With its own SAFE, the accelerator claims $\sigma_a / \min(V_a, \lambda_a(\pi - x))$ of the equity. Multiplied by $\pi - x$, this claim is the larger of the share σ_a/V_a of the equity, when the cap binds, and of the value σ_a/λ_a, when the discount binds. The protection takes at most the whole equity, and leaves in expectation

$$W\Big(x + \frac{\sigma_a}{\lambda_a}\Big) - \frac{\sigma_a}{V_a}\, W\Big(x + \frac{V_a}{\lambda_a}\Big).$$

Proposition 7 (The accelerator's SAFE)**.** *With a capped SAFE for the accelerator, for $\ell > \sigma_a$, the seed investor invests $\sigma_s^* = \ell - \sigma_a$ and takes all the equity the protection leaves, with a priced ($\rho_s = 1$) or an unpriced SAFE. It invests if and only if*

$$\ell - \sigma_a \le W\Big(I + \frac{\sigma_a}{\lambda_a}\Big) - \frac{\sigma_a}{V_a}\, W\Big(I + \frac{V_a}{\lambda_a}\Big),$$

and the accelerator keeps $W(I)$ minus this value, at least $W(I) - W(I + \sigma_a)$, on each of these projects. The accelerator never earns more than with the fixed share. It earns as much when an optimal fixed share (σ_a^P, ρ_a^P) has $\sigma_a^P > 0$ and $\rho_a^P W(I) > W(I) - W(I + \sigma_a^P)$, and comes arbitrarily close to it when $\sigma_a^P = 0$.

→ Proof in Appendix A, page 35

The seed investor invests only its need, so what the protection takes below the scaling cost does not matter, and the accelerator's profit depends only on what it keeps at the threshold I, as with the fixed share. The SAFE only bounds this value from below: it keeps at least $W(I) - W(I + \sigma_a)$, the expected value of $\min(\sigma_a, (\pi - I)^+)$, since it returns its amount when the project allows it. In our simulations, the optimal fixed share keeps more than this bound whenever the accelerator invests (Section 8): capping well is then pricing well.

A fixed share combined with an uncapped SAFE that converts at the equity round gives the accelerator nothing more than the fixed share. With a share ρ_a and such a SAFE of amount u, the accelerator takes $\min(\pi - x, \rho_a(\pi - x) + u)$ in the state $\pi > x$, and leaves $((1 - \rho_a)(\pi - x) - u)^+$, that is, $(1 - \rho_a)W(x + u/(1 - \rho_a))$ in expectation. A dollar more from the seed investor raises this value by $(1 - \rho_a)(1 - F(x + u/(1 - \rho_a))) < 1$, so that the seed investor invests only its need, and the accelerator keeps $W(I) - (1 - \rho_a)W(I + u/(1 - \rho_a))$ on every project the seed investor finances. As with its own SAFE, the accelerator's profit depends only on this value, which the fixed share equal to it divided by $W(I)$ keeps with the same contribution. The combination therefore earns what a fixed share earns, and the fixed share is the combination with $u = 0$.

5 The MFN clause

With an MFN clause, the accelerator keeps at least ρ_a^0, and its amount m converts on the seed investor's terms. Against a priced SAFE of valuation σ_s/ρ_s, its amount converts at $\min(\sigma_s/\rho_s, \pi - x)$. Against an unpriced SAFE, the two SAFEs convert at the same price, $\min(V_s, \lambda(\pi - x))$. For $m = 0$, the clause is a fixed post-money share.

5.1 The seed investor

Proposition 8 (The seed investor's contract)**.** *For $\ell > \sigma_a$, an unpriced SAFE gives the seed investor at least as much as a priced SAFE, and strictly more when $m > 0$. With an unpriced SAFE, the clause caps the seed investor's share at $(1 - \rho_a^0)\sigma_s/(\sigma_s + m)$, which the cap $(\sigma_s + m)/(1 - \rho_a^0)$ attains, and a discount brings it nothing. The seed investor finances the shock if and only if $\ell - \sigma_a + m \leq (1 - \rho_a^0)W(I)$, and invests*

beyond its need if

$$(1-\rho_a^0)\Big[\frac{m\,W(I)}{(\ell-\sigma_a+m)^2}+\frac{(\ell-\sigma_a)\big(1-F(I)\big)}{\ell-\sigma_a+m}\Big]>1.$$

→ Proof in Appendix A, page 36

A priced SAFE keeps its share when the equity round values the project below the seed investor's valuation, while the MFN component converts at the price of this round. The two claims then exceed the equity left by the fixed share, and both lose the same fraction of their claims. An unpriced SAFE converts at the same price as the MFN component in every state, and keeps the share $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ in all of them.

Every dollar the seed investor adds raises its share $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$: it dilutes the MFN component. The gain is large on small rounds, where m weighs a lot against σ_s.

5.2 The accelerator

On a financed project, the accelerator keeps the project's value minus the seed investor's amount and profit. For $\theta=\rho_a^0$ or m, this value has derivative

$$-\frac{\partial\Pi_s}{\partial\theta}(\sigma_s^*)-\Big[F(x^*)+\frac{\partial\Pi_s}{\partial\sigma_s}(\sigma_s^*)\Big]\frac{\partial\sigma_s^*}{\partial\theta},$$

where Π_s is the seed investor's profit: what the clause takes from the seed investor at a given amount, minus the cost of the change in its amount, of which $F(x^*)$ per dollar is lost on the projects that are not fully funded.

Proposition 9. *Take any fixed share (σ_a,ρ_a) with $\sigma_a>0$ and $0<\rho_a<1$. For every small enough $m>0$, the MFN clause with the same contribution, $\rho_a^0=\rho_a$ and the component m earns strictly more. Suppose that the optimal fixed share comes with a positive contribution. Every optimum $(\sigma_a^*,\rho_a^{0*},m^*)$ satisfies:*

- *(i) $m^*>0$: an MFN component does strictly better than any fixed share;*
- *(ii) the seed investor invests beyond its need on the smallest rounds;*
- *(iii) if $\rho_a^{0*}>0$, moving protection from the fixed share to the MFN component at a given last financed project brings a gain, strictly positive on every round without overinvestment, and a cost on the rounds with overinvestment, where every extra dollar is lost on the projects that are not fully funded. The gain equals the cost if $m^*<\sigma_a^*$, and is at least the cost if $m^*=\sigma_a^*$.*

The accelerator earns strictly more than with a fixed share, hence than in the benchmark.

→ Proof in Appendix A, page 39

Some MFN clause beats each such fixed share, not every one: the improvement is shown for small m, and a larger component also moves the last financed project back by m. When the optimal fixed share is the limit with no contribution, an MFN component requires a positive one, and the fixed share may do better.

Point (iii) explains the fixed share. At a given last financed project, the MFN component takes the more from a round the further the round is from the last one: it is a price discrimination device. If the seed investor did not answer, the accelerator would move all of its protection into the clause, until $\rho_a^0 = 0$ or $m = \sigma_a$. The fixed share curbs the seed investor's investment beyond its need, which a stronger fixed protection reduces. When $m^* = \sigma_a^*$, the move still pays, and the constraint $m \leq \sigma_a$ stops it: in our simulations, the optimum lies there in all but one configuration where the accelerator invests (Section 8). The clause covers at most the accelerator's contribution: when the accelerator invests nothing, the MFN component is zero, and the clause is a fixed share.

6 Seed rounds and full funding

The protection changes which projects survive. The probabilities of a seed round and of full funding describe this change; Section 7 measures welfare. The experiment is covered if the accelerator's cash covers the shock, $\ell \leq \sigma_a$, or if a seed investor finances it. The project is fully funded if the experiment is covered and π exceeds the threshold. Its probability of full funding is

$$\int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big) g(\ell)\, d\ell + \int_S \big(1 - F(x^*)\big) g(\ell)\, d\ell,$$

where S is the set of shocks the seed investor finances and x^* the threshold it chooses. A project the accelerator finances alone faces the threshold $I + \ell - \sigma_a \leq I$, since the unused cash goes to the project.

Proposition 10. *Whatever the accelerator's protection and the seed contract, a seed investor finances a shock $\ell > \sigma_a$ only if $\ell - \sigma_a \leq W(I)$. A seed round covers at most $W(I) < I$, and a project is covered only if $\ell \leq \sigma_a + W(I)$.*

→ Proof in Appendix A, page 43

At best, the seed investor takes the whole equity, and every dollar it adds beyond its need is lost on the projects that are not fully funded: its profit is at most $W(I) - (\ell - \sigma_a)$.

Proposition 11. *Consider an MFN clause (ρ_a^0, m) against an unpriced SAFE, which is a fixed share $\rho_a = \rho_a^0$ when $m = 0$, and no protection when $\rho_a^0 = m = 0$. With*

a contribution σ_a, the accelerator covers $[0, \sigma_a]$ alone, and the seed investor finances exactly the shocks

$$\sigma_a < \ell \leq \sigma_a + (1 - \rho_a^0)W(I) - m.$$

On the last one, the accelerator keeps $\rho_a^0 W(I) + m$. The experiment is covered with probability $G\big(\sigma_a + (1 - \rho_a^0)W(I) - m\big)$.

→ Proof in Appendix A, page 43

The seed investor finances a project only if the equity above its amount covers what the accelerator keeps. A protection raises what the accelerator keeps on the last project, and the seed investor stops earlier. A higher contribution moves the whole zone without changing its width, and the probability of a seed round rises or falls depending on whether g is higher at its upper or at its lower end.

Write σ_a^B for the contribution without protection, (σ_a^P, ρ_a^P) for the optimal fixed share, $(\sigma_a^M, \rho_a^{0M}, m^M)$ for an optimal MFN clause, and $x^*(\ell)$ for the threshold the seed investor chooses under the MFN clause.

Proposition 12. *(i) Moving from no protection to the optimal fixed share changes the probability of full funding by*

$$\underbrace{\int_{\sigma_a^B}^{\sigma_a^P} \Big[\int_0^s f(I + \ell - s)\, g(\ell)\, d\ell + \big(1 - F(I)\big) g\big(s + W(I)\big) \Big] ds}_{\text{contribution}} - \underbrace{\big(1 - F(I)\big) \Big[G\big(\sigma_a^P + W(I)\big) - G\big(\sigma_a^P + (1 - \rho_a^P)W(I)\big) \Big]}_{\text{last project} \,>\, 0}.$$

The first term has the sign of $\sigma_a^P - \sigma_a^B$: if the fixed share does not raise the contribution, it strictly lowers the probability of full funding.

(ii) With an MFN clause, on a round where the seed investor invests beyond its need, the project is fully funded with probability $1 - F(x^*) > 1 - F(I)$.

→ Proof in Appendix A, page 43

A dollar more from the accelerator raises the probability of full funding twice: its unused cash lowers the threshold on the projects it finances alone, and the seed investor's zone moves to larger shocks without shrinking. The fixed share leaves the accelerator $\rho_a^P W(I)$ on the last project, and the shocks between $\sigma_a^P + (1 - \rho_a^P)W(I)$ and $\sigma_a^P + W(I)$ are no longer covered.

Proposition 13. *Moving from the optimal fixed share to an optimal MFN clause*

changes the probability of a seed round by

$$\underbrace{\big[G\big(\sigma_a^M+(1-\rho_a^P)W(I)\big)-G(\sigma_a^M)\big]-\big[G\big(\sigma_a^P+(1-\rho_a^P)W(I)\big)-G(\sigma_a^P)\big]}_{\textit{contribution}}$$
$$+\underbrace{G\big(\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M\big)-G\big(\sigma_a^M+(1-\rho_a^P)W(I)\big)}_{\textit{last project}},$$

and the probability of full funding by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M}\Big[\int_0^s f(I+\ell-s)\,g(\ell)\,d\ell+\big(1-F(I)\big)g\big(s+(1-\rho_a^P)W(I)\big)\Big]ds}_{\textit{contribution}}$$
$$+\underbrace{\big(1-F(I)\big)\Big[G\big(\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M\big)-G\big(\sigma_a^M+(1-\rho_a^P)W(I)\big)\Big]}_{\textit{last project}}$$
$$+\underbrace{\int_{\sigma_a^M}^{\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M}\big(F(I)-F(x^*(\ell))\big)g(\ell)\,d\ell}_{\textit{investment beyond the need}\,\geq\,0}.$$

→ Proof in Appendix A, page 44

The last-project terms have the sign of $\rho_a^P W(I)-\rho_a^{0M}W(I)-m^M$, the difference in what the accelerator keeps on the last project, the contribution term of full funding the sign of $\sigma_a^M-\sigma_a^P$, and the contribution term of the seed round no sign. Proposition 9 determines none of them; only the investment beyond the need has a sign.

The investment beyond the need acts on the accelerator's choices. Fix a contribution σ_a and an MFN clause (ρ_a^0, m). On a shock ℓ, write $\Pi_s^*(\ell)$ for the seed investor's profit; it depends on ℓ only through $\ell-\sigma_a$, and so does $x^*(\ell)$.

Proposition 14. *(i) The accelerator's profit is*

$$B(\sigma_a)+\int_{\sigma_a}^{\sigma_a+(1-\rho_a^0)W(I)-m}\Big[W(I)-(\ell-\sigma_a)-\Pi_s^*(\ell)-\int_{x^*(\ell)}^I F(x)\,dx\Big]g(\ell)\,d\ell.$$

(ii) Suppose that the optimal fixed share at σ_a finances the same last shock, $\sigma_a+(1-\rho_a^0)W(I)-m$. The accelerator gains from lowering ρ_a^0, which extends the seed investor's zone, if and only if

$$\int_{\sigma_a}^{\sigma_a+(1-\rho_a^0)W(I)-m}\Big[W(I)+\frac{\partial}{\partial\rho_a^0}\Big(\Pi_s^*(\ell)+\int_{x^*(\ell)}^I F(x)\,dx\Big)\Big]g(\ell)\,d\ell>0.$$

On a round where the seed investor invests only its need, the integrand is $mW(I)/(\ell-\sigma_a+m)>0$. On a round where it invests more, the integrand contains $F(x^*)\,\partial\sigma_s^*/\partial\rho_a^0<0$.

(iii) With $\bar{\ell} = \sigma_a + (1-\rho_a^0)W(I) - m$ the last shock the seed investor finances, the derivative of the profit in σ_a is

$$\big(\rho_a^0 W(I) + m\big)g(\bar{\ell}) - \big(1 - G(\bar{\ell})\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)\, g(\ell)\, d\ell \\ - \int_{\sigma_a}^{\bar{\ell}} \Big[\Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(x)\, dx\Big] g'(\ell)\, d\ell.$$

→ Proof in Appendix A, page 44

By (i), the accelerator keeps, on each shock the seed investor finances, the value of the project at the threshold I net of the round, minus the seed investor's profit, minus the surplus lost by the investment beyond the need, $\int_{x^*}^{I} F(x)\, dx$: each extra dollar lowers the threshold, and is lost on the projects that are not fully funded, with probability $F(x)$ at threshold x.

Point (ii) is the last-project channel. The MFN clause takes more from small rounds than a fixed share with the same last shock: on a round without investment beyond the need, a stronger protection lowers the seed investor's profit by less than $W(I)$, and the accelerator gains from extending the zone. The investment beyond the need pulls the other way: a stronger protection curbs it, which recovers lost surplus, and the accelerator gains from protecting more and covering fewer shocks.

Point (iii) is the contribution channel. The accelerator pays an extra dollar of contribution on every shock. On the projects it finances alone, the dollar comes back except on those that are not fully funded, and on the shocks no one covers, it is lost. On a shock the seed investor finances, the dollar lowers by one dollar the round the seed investor must bring, $\ell - \sigma_a$, and the accelerator recovers only what a smaller round leaves it. With a fixed share, $m = 0$, the seed investor invests its need, and its profit, $(1-\rho_a)W(I) - (\ell - \sigma_a)$, rises by the whole dollar: the accelerator keeps $\rho_a W(I)$ whatever the round. With the MFN clause, on a round without investment beyond the need, the accelerator keeps $\rho_a^0 W(I) + (1-\rho_a^0)mW(I)/(\ell - \sigma_a + m)$, and recovers $(1-\rho_a^0)mW(I)/(\ell - \sigma_a + m)^2$ of the dollar: the component weighs more against a smaller round. In the expression of (iii), the term $1 - G(\bar{\ell})$ counts the dollar as lost only on the shocks no one covers, and the last integral takes back what the seed investor's profit and the lost surplus gain from the smaller rounds. The MFN clause lowers the seed investor's profit most on small rounds, and the investment beyond the need adds lost surplus on these same rounds. Whether the accelerator invests more or less than with a fixed share depends on these two profiles and on the slope of g over the zone.

7 Total surplus

The probability of full funding counts every fully funded project the same, whatever its value. The total surplus is the sum of the payoffs of the accelerator, the seed investor and the equity investor, whose profit is zero. The transfers between them cancel: on a covered project with threshold x, the total surplus is the value of the equity minus all the money brought before the equity round, $W(x) - (I + \ell - x)$. On a project abandoned for lack of a seed round, it is $-\sigma_a$, the contribution spent. Since $W' = -(1 - F)$, the surplus of a covered project equals

$$W(I) - \ell - \int_x^I F(t)\, dt.$$

The last term is the cost of the money brought beyond the need, $I - x$. Integrating by parts,

$$\int_x^I F(t)\, dt = (I - x)\, F(x) + \int_x^I (I - \pi)\, f(\pi)\, d\pi :$$

this money is lost on the projects that are abandoned, $\pi < x$, and finances projects worth less than the scaling cost, $x < \pi < I$.

The surplus of a covered project is at most $W(I) - \ell$, reached at the threshold I, and the surplus of an abandoned project is at most zero. The first best contributes nothing and covers exactly the shocks $\ell \leq W(I)$, at the threshold I:

$$TS^* = \int_0^{W(I)} \big(W(I) - \ell\big)\, g(\ell)\, d\ell.$$

Proposition 15. *With an MFN clause (ρ_a^0, m), which covers the fixed share and the absence of protection (Proposition 11), let $\bar{\ell} = \sigma_a + (1 - \rho_a^0)W(I) - m$ be the last shock the seed investor finances. The total surplus is*

$$TS = TS^* - L_{acc} - L_{cov} - L_{over} - L_{sunk},$$

where

$$L_{acc} = \int_0^{\sigma_a} \int_{I+\ell-\sigma_a}^{I} F(t)\, dt\; g(\ell)\, d\ell, \qquad L_{cov} = \int_{\bar{\ell}}^{W(I)} \big(W(I) - \ell\big)\, g(\ell)\, d\ell \geq 0,$$

$$L_{over} = \int_{\sigma_a}^{\bar{\ell}} \int_{x^*(\ell)}^{I} F(t)\, dt\; g(\ell)\, d\ell, \qquad L_{sunk} = \sigma_a\big(1 - G(\bar{\ell})\big).$$

When the seed investor invests only its need, a dollar more of contribution, at a given amount kept on the last project, changes the total surplus by

$$\big(\rho_a^0 W(I) + m\big) g(\bar{\ell}) - \big(1 - G(\bar{\ell})\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)\, g(\ell)\, d\ell,$$

and a dollar more kept on the last project, at a given contribution, by $-(\rho_a^0 W(I) + m)\, g(\bar{\ell})$.

→ Proof in Appendix A, page 46

L_{acc} is the cost of the money the accelerator brings beyond the shock, L_{over} the cost of the money the seed investor brings beyond its need, and L_{sunk} the contribution spent on the abandoned projects. L_{cov} is a loss in both directions: if $\bar{\ell} < W(I)$, the shocks with positive NPV that the seed investor no longer covers; if $\bar{\ell} > W(I)$, the shocks with negative NPV that it covers because the accelerator's contribution is already spent.

Without protection, the accelerator keeps nothing on the last project, and the total surplus falls with the contribution. The marginal gain of the accelerator in Proposition 3, $W(I)\,g(\sigma_a)$, is a project it takes from the seed investor, who would have financed it at the threshold I. The accelerator invests whenever $\sigma_a^B > 0$, while the total surplus is maximal at $\sigma_a = 0$.

Proposition 16. *Moving from no protection to the optimal fixed share changes the total surplus by*

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^B} \Big[\big(1 - G(s + W(I))\big) + \int_0^s F(I + \ell - s)\, g(\ell)\, d\ell\Big] ds}_{\textit{contribution}} - \underbrace{\int_{\sigma_a^P + (1-\rho_a^P)W(I)}^{\sigma_a^P + W(I)} \big(\sigma_a^P + W(I) - \ell\big)\, g(\ell)\, d\ell}_{\textit{last project} \,>\, 0}.$$

The first term has the sign of $\sigma_a^B - \sigma_a^P$.

→ Proof in Appendix A, page 46

Proposition 17. *Moving from the optimal fixed share to an optimal MFN clause changes the total surplus by*

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M} \Big[\rho_a^P W(I)\, g\big(s + (1-\rho_a^P)W(I)\big) - \big(1 - G(s + (1-\rho_a^P)W(I))\big) - \int_0^s F(I + \ell - s)\, g(\ell)\, d\ell\Big] ds}_{\textit{contribution}} + \underbrace{\int_{\sigma_a^M + (1-\rho_a^P)W(I)}^{\sigma_a^M + (1-\rho_a^{0M})W(I) - m^M} \big(\sigma_a^M + W(I) - \ell\big)\, g(\ell)\, d\ell}_{\textit{last project}} - \underbrace{\int_{\sigma_a^M}^{\sigma_a^M + (1-\rho_a^{0M})W(I) - m^M} \int_{x^*(\ell)}^{I} F(t)\, dt\; g(\ell)\, d\ell}_{\textit{investment beyond the need} \,\geq\, 0}.$$

→ Proof in Appendix A, page 47

The last-project term has the sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$, and the contribution term no sign. On each shock the seed investor finances in addition, the total surplus gains $\sigma_a + W(I) - \ell$, the value of the project at the threshold I net of the round, against $-\sigma_a$ if it were abandoned. The investment beyond the need raises the probability of full funding (Proposition 12) and lowers the total surplus: the projects it brings above the threshold are worth less than I.

8 Software and deep tech

We sort projects by the size of their liquidity needs relative to their scaling cost, $E(\ell)/I$. A software-like project has a small ratio: its need is mostly the scaling cost. A deep-tech project has a ratio above one: its need comes mostly as liquidity shocks.

8.1 Setting

The model is homogeneous of degree one in $(\pi, I, \ell, \sigma_a, m, \sigma_s)$, so we normalize $I = 1$. We take a uniform f on $[0, 1.8]$, so that $E(\pi) = 0.9 < I$ and $W(I) \approx 0.178$. For ℓ, we take a Gamma distribution with shape 2, which is log-concave, with mean $E(\ell)$ from $0.015\,I$ to $4I$. Of the shapes we tried, it is the most skewed. Section 9 shows that a Gamma with shape 5 and a Weibull with shape 3 give the same results. The benchmark contribution maximizes B on a grid. For the fixed share, $m = 0$, and for the MFN clause, each imposed m/σ_a on a grid, we maximize over σ_a and ρ_a^0, the seed investor's answer being computed round by round with an unpriced SAFE.

8.2 Contracts

The configurations split into two regimes (Figures 2 and 3). For software-like projects, $E(\ell)/I \leq 0.05$, the accelerator invests: 0.04 to $0.10\,I$ without protection, less with the fixed share, and less again with the MFN clause. All of its money goes through the clause, $m^*/\sigma_a^* = 1$. The MFN clause gives the highest profit (Figure 1): it raises the accelerator's profit by 1.5% and 5.4% over the optimal fixed share, at $E(\ell)/I = 0.015$ and 0.05. When the accelerator invests, the optimal fixed share keeps 0.125 to 0.152 per seed project, well above $W(I) - W(I + \sigma_a^P) \leq 0.027$, what a capped SAFE of the same contribution keeps at least: the SAFE would do as well. From $E(\ell)/I = 0.15$ on, the accelerator's optimal contribution vanishes under any protection: few shocks are small enough for it to cover them alone, and on all the others its money goes to the seed investor or is lost. The MFN component is then zero, since $m \leq \sigma_a$, and the protection is a fixed share between 0.34 and 0.45, bought with an amount and at a

valuation that tend to zero. The MFN clause then gives the same profit as the fixed share, 0.45 to 0.52 of the first-best surplus, against nothing without protection.

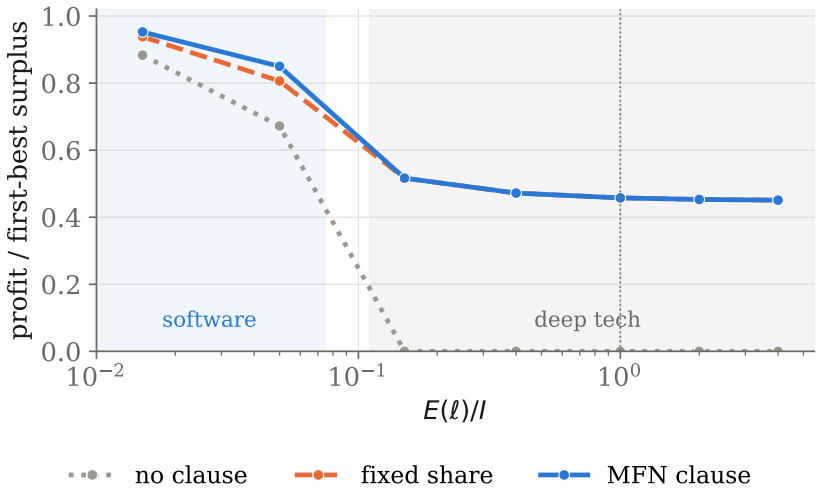


Figure 1: Accelerator's profit relative to the first-best surplus, without protection, at the optimal fixed share and at the optimal MFN clause. Gamma(2) shocks.

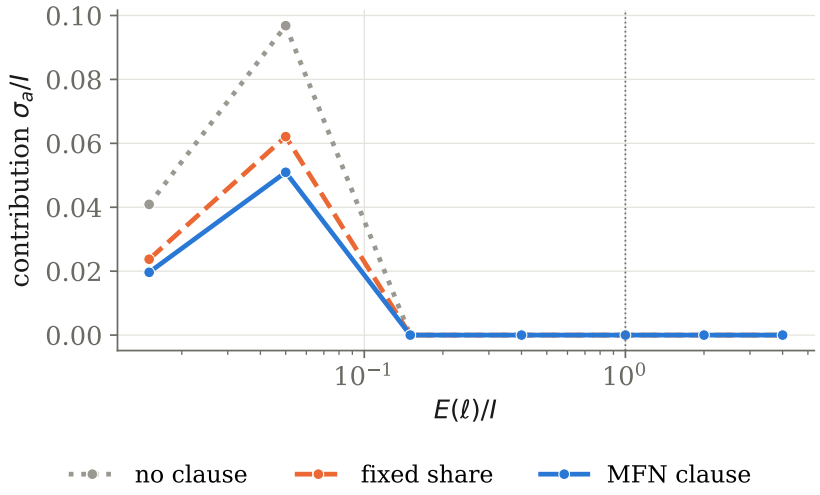


Figure 2: Accelerator's contribution σ_a/I without protection, at the optimal fixed share and at the optimal MFN clause. Gamma(2) shocks.

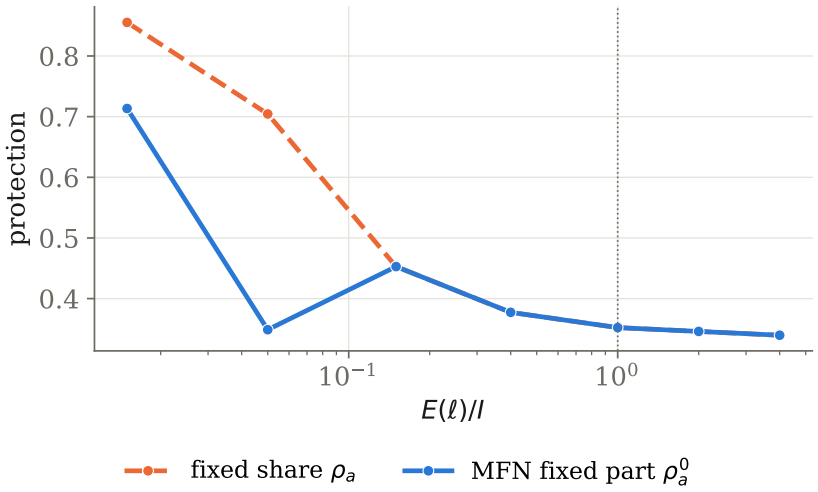


Figure 3: Optimal fixed share ρ_a^* and fixed part of the optimal MFN clause ρ_a^{0*}. Gamma(2) shocks.

8.3 Total surplus

For software-like projects, the MFN clause gives a higher total surplus than the fixed share, which gives a higher one than no protection: 0.88–0.96 of the first best with the clause, 0.86–0.96 with the fixed share, 0.79–0.91 without protection (Figure 4). The protections lower the contribution, which the accelerator brings beyond what maximizes total surplus. From $E(\ell)/I = 0.15$ on, the absence of protection reaches the first best: the accelerator invests nothing, and the seed investor, which takes the whole project, covers exactly the shocks below $W(I)$ at the threshold I. The protection removes the shocks between $(1-\rho_a)W(I)$ and $W(I)$, and the total surplus falls to 0.74–0.77 of the first best.

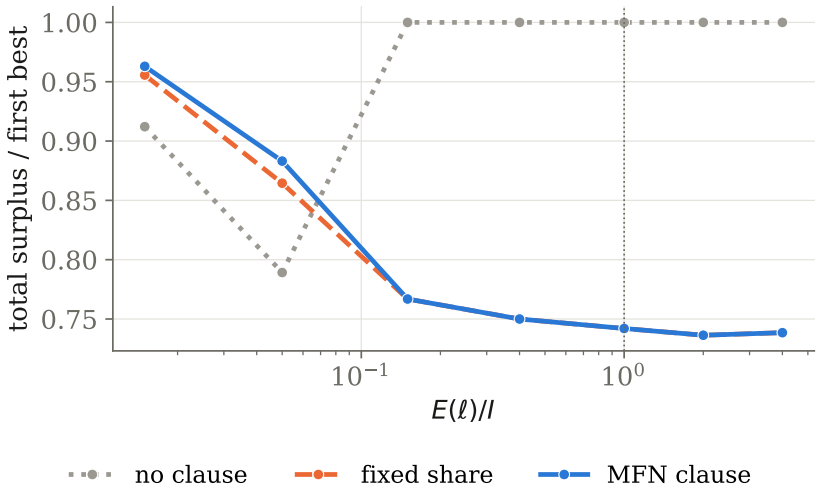


Figure 4: Total surplus relative to the first best. Gamma(2) shocks.

8.4 Seed rounds and full funding

The probabilities show which projects survive. For software-like projects, protection raises the probability of a seed round, from 0.03–0.10 without protection to 0.17–0.23 with the fixed share and 0.26–0.34 with the MFN clause, since the accelerator invests less and leaves more shocks to the seed investor. It slightly lowers the probability of full funding, from 0.46–0.47 to 0.43–0.45 (Figures 5 and 6). From $E(\ell)/I = 0.15$ on, the accelerator's contribution vanishes, and the protection only shortens the seed investor's zone: it roughly halves both probabilities. At $E(\ell)/I = 0.15$, the probability of a seed round falls from 0.69 to 0.37, and that of full funding from 0.30 to 0.17. By Proposition 10, both probabilities vanish as the shocks move above $W(I)$.

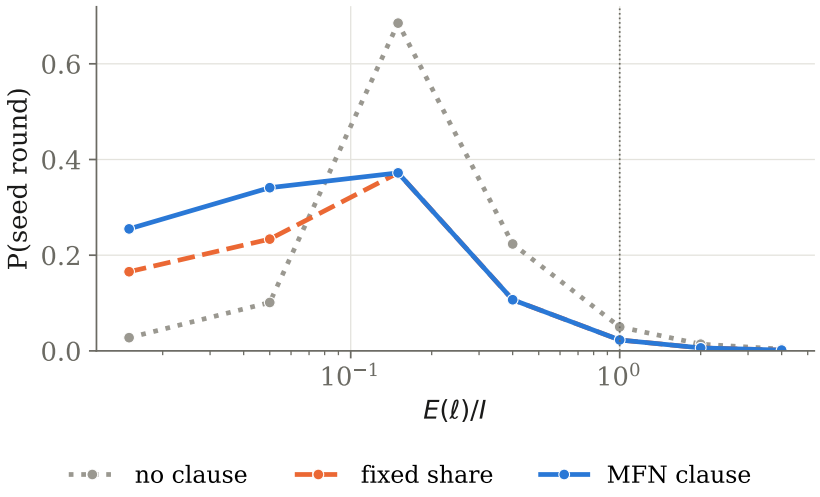


Figure 5: Probability of a seed round. Gamma(2) shocks.

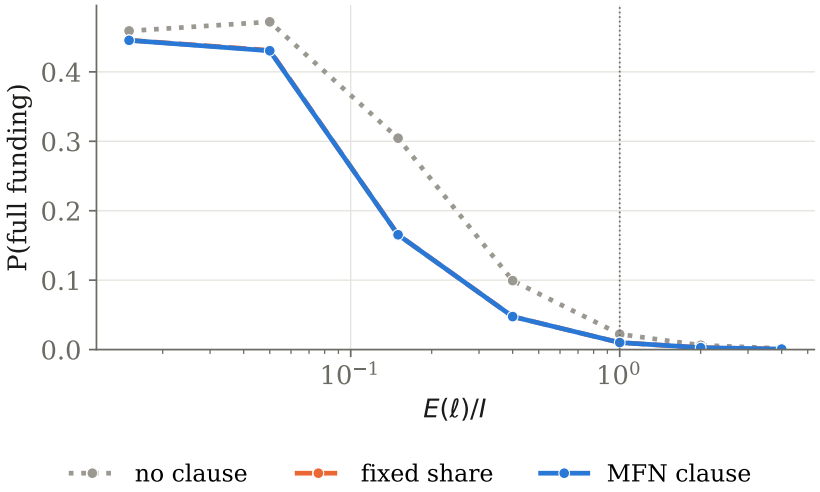


Figure 6: Probability of full funding. Gamma(2) shocks.

9 Robustness and extensions

9.1 Other distributions of the liquidity shock

With a Gamma of shape 5 or a Weibull of shape 3 instead of a Gamma of shape 2, the two regimes and the ranking of the protections are unchanged (Figure 7). For software-like projects, the accelerator invests 0.03 to 0.09 I without protection, m^*/σ_a^* lies between 0.95 and 1, and the MFN clause gives the highest profit, 1 to

4% above the fixed share, and the highest total surplus: 0.91–0.98 of the first best with the clause, 0.90–0.98 with the fixed share, 0.82–0.95 without protection. From $E(\ell)/I = 0.15$ on, the accelerator's contribution vanishes, the fixed share lies between 0.18 and 0.33, and the total surplus falls to 0.73–0.76 of the first best.

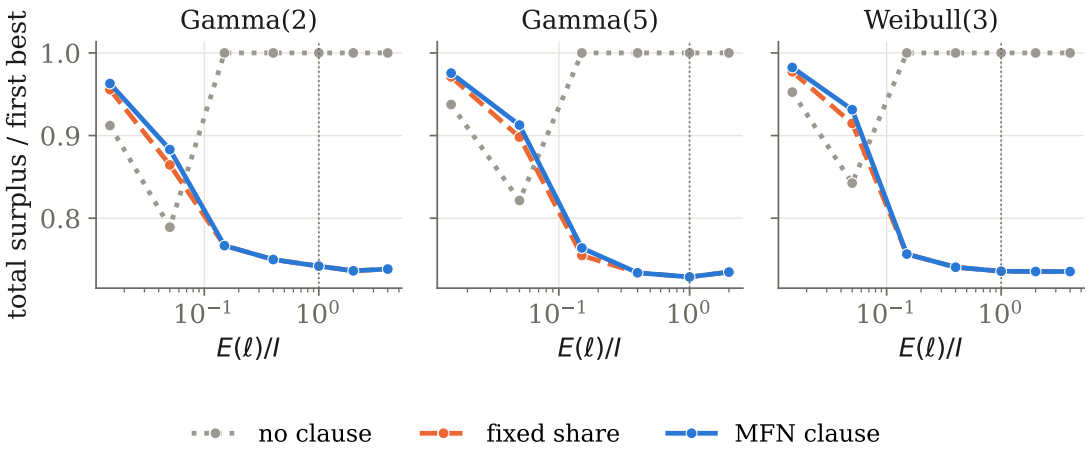


Figure 7: Total surplus relative to the first best, for the three distributions of the liquidity shock.

9.2 Another distribution of the project's value

We replace the uniform f by a Beta$(2, 2)$ distribution stretched over $[0, 1.8]$, which is log-concave, has the same mean $E(\pi) = 0.9$, and gives $W(I) \approx 0.123$. The two regimes are unchanged. For software-like projects, the MFN clause gives the highest profit, $m^*/\sigma_a^* = 0.95$ to 1, and its fixed part lies between 0.22 and 0.76; the total surplus is 0.82–0.97 of the first best with the clause, 0.80–0.97 with the fixed share and 0.71–0.93 without protection. From $E(\ell)/I = 0.15$ on, the accelerator's contribution vanishes, its fixed share lies between 0.18 and 0.42, and it lowers the total surplus to 0.74–0.76 of the first best, which the absence of protection reaches.

9.3 A priced SAFE against the MFN clause

The seed investor prefers the unpriced SAFE (Proposition 8). If the accelerator could require a priced SAFE, the MFN clause would still beat any fixed share.

Proposition 18. *Against a seed investor that uses a priced SAFE, take any fixed share (σ_a, ρ_a) with $\sigma_a > 0$ and $0 < \rho_a < 1$. For every small enough $m > 0$, the MFN clause with the same contribution, $\rho_a^0 = \rho_a$ and the component m earns strictly more. If the optimal fixed share comes with a positive contribution, every optimum has $m^* > 0$, and the accelerator earns strictly more than with any fixed share.*

→ Proof in Appendix A, page 47

The seed investor's share never exceeds $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$, so a small MFN component raises the value the accelerator keeps on every round by at least $(1-\rho_a^0)\,m\,W(I)/(\ell-\sigma_a+m)$, and the seed investor stops financing only rounds within $O(m)$ of the ends of its zone.

9.4 The accelerator's own SAFE as the fixed part

The fixed part of the clause can also be the accelerator's own SAFE, on the rest of its contribution, $\sigma_a - m$. The seed investor and the MFN component then share what this SAFE leaves, and the fixed share is the limit of a SAFE of cap $V_a = (\sigma_a - m)/\rho_a^0$ without discount as m tends to σ_a. The value $(\sigma_a - m)/\lambda_a$ that the SAFE keeps in the low states changes the seed investor's answer only on the small rounds where it invests beyond its need. We compute the optimal clause with this fixed part, in the setting of Section 8. It earns at most $2 \cdot 10^{-5}$ more than the clause with a fixed share, within numerical precision: the value the SAFE keeps in the low states is worth nothing to the accelerator.

Y Combinator's 7% took both forms. From 2019, its $150,000 for 7% was a post-money SAFE whose cap gave it at least 7%, without MFN component, the capped SAFE of Section 4.[8] Its current $125,000 "converts into a fixed 7%", a fixed share.[9]

9.5 The founders' stake

So far the entrepreneur sells the whole project. Suppose instead that the founders keep a share ρ_e of the equity held before the equity round, which they need to keep working on the project. Their stake is written before the seed round, and comes first, with the accelerator's protection: the seed investor's share may not exceed $1-\rho_e-\rho_a$ with a fixed share, and $1-\rho_e$ without protection. The accelerator's own SAFE comes after the founders' stake, so that its claim is limited to $1-\rho_e$. On the shocks it covers alone, the accelerator keeps $(1-\rho_e)W(I+\ell-\sigma_a)$.

Proposition 19 (The founders' stake)**.** *With a founders' stake ρ_e:*

(i) the seed investor's answers of Propositions 1, 4, 5 and 8 hold with ρ_a replaced by $\rho_e+\rho_a$, ρ_a^0 by $\rho_e+\rho_a^0$, and no protection by a fixed share ρ_e. Without protection, the seed investor finances the shock if and only if $\ell-\sigma_a \le (1-\rho_e)W(I)$, with the cap $\sigma_s^*/(1-\rho_e) > \sigma_s^*$, and with $\sigma_a = 0$ the total surplus is

$$TS^* - \int_{(1-\rho_e)W(I)}^{W(I)} \big(W(I)-\ell\big)\, g(\ell)\, d\ell < TS^*;$$

[8] Michael Seibel, "New Standard Deal," Y Combinator blog, September 28, 2018.
[9] The Y Combinator Deal.

(ii) a fixed share ρ_a leaves the accelerator $\rho_a W(I)$ on every project the seed investor finances, and the optimal fixed share earns strictly more than no protection;

(iii) against the accelerator's own capped SAFE, the seed investor invests its need, and finances the shock if and only if $V_a \geq \sigma_a/(1-\rho_e)$ and

$$\ell - \sigma_a \leq (1-\rho_e)\, W\Big(I + \frac{\sigma_a}{\lambda_a(1-\rho_e)}\Big) - \frac{\sigma_a}{V_a}\, W\Big(I + \frac{V_a}{\lambda_a}\Big).$$

The accelerator never earns more than with the fixed share. It earns as much when an optimal fixed share (σ_a^P, ρ_a^P) has $\sigma_a^P > 0$ and $\rho_a^P W(I) > (1-\rho_e)\big[W(I) - W(I + \sigma_a^P/(1-\rho_e))\big]$, and comes arbitrarily close to it when $\sigma_a^P = 0$;

(iv) Propositions 9 (i) and 18 hold: if the optimal fixed share comes with a contribution, every optimal MFN clause has $m^* > 0$, and the accelerator earns strictly more than with any fixed share.

→ Proof in Appendix A, page 49

The seed investor's cap now exceeds its amount, and the absence of protection no longer reaches the first best: the founders' stake shortens the seed investor's zone, as a fixed share does. The ranking of the protections for the accelerator does not depend on the founders' stake, and the stake changes only the bound below which the accelerator's SAFE does as well as the fixed share.

We compute the optimal contracts with $\rho_e = 0.25$ and $\rho_e = 0.5$, the uniform f, and $E(\ell)/I$ from 0.015 to 1. With $\rho_e = 0.25$, the two regimes are unchanged. For software-like projects, the MFN clause gives the highest profit, with $m^* = \sigma_a^*$ or nearly, and a total surplus of 0.86–0.98 of the first best, against 0.85–0.98 with the fixed share and 0.85–0.96 without protection. From $E(\ell)/I = 0.15$ on, the accelerator's contribution vanishes, the absence of protection gives 0.59–0.94 of the first best, and the fixed share roughly halves it, to 0.30–0.62.

With $\rho_e = 0.5$, a third regime appears at $E(\ell)/I = 0.05$ with Gamma shocks. The optimal fixed share comes with a contribution that tends to zero, while the optimal MFN clause keeps the accelerator investing, $\sigma_a^* = 0.03$ to 0.04, all of it through the clause. The MFN clause gives 0.79–0.87 of the first best, against 0.62–0.67 with the fixed share and 0.90–0.92 without protection. The boundary between the regimes therefore depends on the protection, and not only on $E(\ell)/I$.

10 Discussion

The accelerator needs a protection because the seed investor has market power once the shock is realized. If seed investors competed and earned zero profit, the seed investor would leave the accelerator all the surplus it can take, and no protection would have value. With market power, the MFN clause lowers the seed investor's

profit at a given protection, but the comparison at the accelerator's optimal protection is open, since the clause also changes the set of projects the seed investor finances.

In the model, the founders sell the whole project to the accelerator in exchange for the experiment its program offers, and the accelerator's contribution has value only on the shocks it covers alone: on the other shocks, the seed investor takes it with the project, or it is lost. A contribution that also bought a stake from the founders would have a value on every project, and would let the MFN clause operate where, in our simulations, the accelerator's contribution vanishes. The founders' stake of Section 9 goes in this direction, since, when the founders keep half of the equity, the clause keeps the accelerator investing where a fixed share does not.

For large liquidity needs, the accelerator's optimal contract brings no money and keeps a claim that the seed investor cannot dilute, and such accelerators exist. Newchip, an online accelerator, charged the startups of its programs fees from a few thousand dollars to about $20,000, and took in each a warrant, "the right to buy $250,000 worth of shares in the company at a later date".[10] A warrant is an option rather than a share, and the fees are a negative contribution, which the model excludes. Newchip filed for bankruptcy in March 2023, and the court ordered the auction of its warrants in more than 1,000 startups. In the model, a protection without money lowers the total surplus of these projects to about three quarters of the first best.

We compare fixed shares, the accelerator's own SAFE and MFN clauses, the protections that accelerators use. The optimal protection among all contracts is not characterized, since the shock is not contractible and the seed investor answers through its amount, and we leave it, together with equity rounds with pro rata rights and startups that raise several seed rounds, for further research.

11 Conclusion

We model the accelerator's protection against a seed investor with market power, and use the model to evaluate the MFN clause that Y Combinator added to its standard deal in 2022.

We summarize. First, without protection, the seed investor takes all of the equity of the projects it finances, with a priced SAFE, a capped SAFE, or an uncapped SAFE whose discount it chooses; it never uses an uncapped SAFE without discount. The accelerator, whose contribution is spent, keeps value only on the shocks it covers alone. Second, a fixed post-money share gives the accelerator this share of every project the seed investor finances, priced and unpriced SAFEs give the same outcome, and a capped SAFE for the accelerator does no better than the fixed share. Third, every fixed share that comes with a contribution is strictly beaten by an MFN clause with

[10] TechCrunch, "This Austin accelerator made big claims," May 19, 2023, and "They thought they were joining an accelerator. Instead, they lost their startups," May 2, 2024.

the same contribution and a small MFN component, against a priced or an unpriced SAFE, because the component discriminates between rounds; against the clause, the seed investor prefers the unpriced SAFE, invests beyond its need in the smallest rounds, and the fixed part of the clause curbs this answer, while the accelerator's contribution bounds the clause. Fourth, a seed round covers at most the project's expected surplus at the scaling cost, which is below the scaling cost. The protections change the probabilities of a seed round and of full funding through the contribution and through what the accelerator keeps on the last project, and the investment beyond the need, which raises the probability of full funding, lowers the total surplus. Fifth, numerically, the accelerator invests and puts almost all of its money through the clause only when liquidity needs are small relative to the scaling cost; for larger needs, its contribution vanishes, and its fixed share lowers the total surplus to 0.74–0.77 of the first best, which the absence of protection reaches.

The optimal contracts take two forms. For software-like projects, all of the accelerator's money converts through the MFN component, and the fixed part of the clause, between 0.35 and 0.71, is bought on top of it with an amount and at a valuation that tend to zero, as a priced or as a capped SAFE. For projects whose liquidity needs are larger, the accelerator buys a fixed share between 0.34 and 0.45 in the same way, which the seed investor's terms cannot dilute.

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A Proofs

Throughout, $x = I + \ell - \sigma_a - \sigma_s$, and since $\partial x/\partial \sigma_s = -1$, the equity value $W(x)$ has derivative $1 - F(x)$ in σ_s. By Leibniz, $W'(x) = -(1 - F(x))$: the boundary term vanishes because the integrand is zero at $\pi = x$.

Proof of Proposition 1

The seed investor solves

$$\max_{\sigma_s,\, \rho_s} \Pi_s = \rho_s\, W(x) - \sigma_s$$

subject to (L) $\sigma_s \geq \ell - \sigma_a$ and $\rho_s \in [0, 1]$. The accelerator accepts any offer, since rejecting it gives it nothing.

We find the share first. Π_s strictly increases in ρ_s as soon as $W(x) > 0$: the seed investor takes $\rho_s = 1$, and $\Pi_s = W(x) - \sigma_s$.

(L) binds. This profit has derivative $(1 - F(x)) - 1 = -F(x) \leq 0$ in σ_s. One more unit of σ_s costs one unit and is returned only in the states where the project is fully funded. The seed investor goes down to $\sigma_s = \ell - \sigma_a$, where $x = I$, and earns $W(I) - (\ell - \sigma_a)$. If this is negative, no larger amount does better. The seed investor therefore invests if and only if $\ell - \sigma_a \leq W(I)$, and the accelerator keeps $(1 - \rho_s^*)W(I) = 0$. □

Proof of Proposition 2

(i) In state $\pi > x$, the SAFE receives $\min(\sigma_s/(\pi - x), 1)(\pi - x) = \min(\sigma_s, \pi - x)$. We show that the seed investor loses money at every amount. Its gross payoff is $\pi - x$ on $x < \pi < x + \sigma_s$ and σ_s above, with $F = 1$ above 1:

$$\int_x^{x+\sigma_s} (\pi - x)\, f(\pi)\, d\pi + \sigma_s\big(1 - F(x + \sigma_s)\big).$$

Integrating the first term by parts, it equals $\sigma_s F(x + \sigma_s) - \int_x^{x+\sigma_s} F(\pi)\, d\pi$, and the gross payoff is $\sigma_s - \int_0^{\sigma_s} F(x + t)\, dt$. Hence $\Pi_s = -\int_0^{\sigma_s} F(x + t)\, dt$. Since $x \geq 0$ and $f > 0$, $F(x + t) > 0$ for every $t > 0$, and $\Pi_s < 0$. The project is abandoned, and the accelerator keeps nothing.

(ii) The seed investor's gross payoff is at most the equity, $W(x)$, so its profit is at most $W(x) - \sigma_s$, which has derivative $-F(x) \leq 0$ in σ_s: it is at most $W(I) - (\ell - \sigma_a)$, the profit of Proposition 1. Take $\sigma_s = \ell - \sigma_a$, so that $x = I$.

With an uncapped SAFE with discount, the SAFE receives $\min\big((\ell - \sigma_a)/\lambda, \pi - I\big)$. With $\lambda = \min\big(1, (\ell - \sigma_a)/(1 - I)\big)$, the face value $(\ell - \sigma_a)/\lambda$ is at least $1 - I \geq \pi - I$, and the SAFE receives the whole equity in every state.

With a cap $V_s \le \ell - \sigma_a$ and any discount, the claim $(\ell - \sigma_a)/\min(V_s, \lambda(\pi - I))$ is at least $(\ell - \sigma_a)/V_s \ge 1$ in every state, and the SAFE receives the whole equity.

In both cases, the seed investor attains $W(I) - (\ell - \sigma_a)$, invests if and only if $\ell - \sigma_a \le W(I)$, and the accelerator keeps nothing. □

Proof of Proposition 3

We show first that an optimum exists. B is continuous and $B(0) = 0$. For $\sigma_a \ge I$, $W(I + \ell - \sigma_a) \le W(I - \sigma_a) = E(\pi) - I + \sigma_a$, so $B(\sigma_a) \le G(\sigma_a)(E(\pi) - I + \sigma_a) - \sigma_a < 0$ by (A1). The maximum is reached on $[0, I)$.

We derive next the first-order condition. By Leibniz, the boundary term evaluates the integrand at $\ell = \sigma_a$, where the threshold is I, and the inner term uses $\partial W(I + \ell - \sigma_a)/\partial \sigma_a = 1 - F(I + \ell - \sigma_a)$:

$$\begin{aligned} B'(\sigma_a) &= W(I)\, g(\sigma_a) + \int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big) g(\ell)\, d\ell - 1 \\ &= W(I)\, g(\sigma_a) - \big(1 - G(\sigma_a)\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)\, g(\ell)\, d\ell. \end{aligned}$$

At a positive optimum, $B'(\sigma_a^B) = 0$, which is the stated condition. □

Proof of Proposition 4

The accelerator holds a post-money share ρ_a, which the seed round does not dilute. The seed investor solves

$$\max_{\sigma_s,\, \rho_s} \ \Pi_s = \rho_s\, W(x) - \sigma_s$$

subject to (L) $\sigma_s \ge \ell - \sigma_a$ and (PM) $\rho_s \le 1 - \rho_a$.

We find the share first. Π_s strictly increases in ρ_s as soon as $W(x) > 0$: (PM) binds, $\rho_s = 1 - \rho_a$.

We find the amount next. $(1 - \rho_a)W(x) - \sigma_s$ has derivative $(1 - \rho_a)(1 - F(x)) - 1 < 0$ in σ_s. Hence $\sigma_s^* = \ell - \sigma_a$ and $x^* = I$. The seed investor earns $(1 - \rho_a)W(I) - (\ell - \sigma_a)$, invests if and only if this is nonnegative, and the accelerator keeps $\rho_a W(I)$. □

Proof of Proposition 5

By the contracts of Section 2, the seed investor receives, in state π, the share

$$\min\Big(\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))},\ 1 - \rho_a\Big)$$

of the equity $\pi - x$: its claim, capped by the fixed share.

We bound first the seed investor's profit. Its share never exceeds $1 - \rho_a$, so its profit is at most $(1 - \rho_a)W(x) - \sigma_s$, which has derivative $(1 - \rho_a)(1 - F(x)) - 1 < 0$ in σ_s. The bound is largest at $\sigma_s = \ell - \sigma_a$, where $x = I$: $\Pi_s \leq (1 - \rho_a)W(I) - (\ell - \sigma_a)$, the profit of Proposition 4.

We show next that this bound is attained. Take $\sigma_s = \ell - \sigma_a$ and a cap $V_s \leq (\ell - \sigma_a)/(1 - \rho_a)$. With any discount,

$$\frac{\ell - \sigma_a}{\min(V_s, \lambda(\pi - I))} \geq \frac{\ell - \sigma_a}{V_s} \geq 1 - \rho_a,$$

so the share equals $1 - \rho_a$ in every state $\pi > I$, and the seed investor attains the bound. A discount brings nothing. □

Proof of Proposition 6

By Propositions 4 and 5, the program is the same against a priced and an unpriced SAFE.

(i) At given σ_a, the term $\rho_a W(I)[G(\sigma_a + (1 - \rho_a)W(I)) - G(\sigma_a)]$ vanishes at $\rho_a = 0$ and at $\rho_a = 1$, where the seed investor finances nothing, and is strictly positive in between, since $g > 0$. Hence ρ_a^* is interior and the derivative in ρ_a vanishes; dividing by $W(I)$ gives the stated condition. On the left, what a stronger protection returns on each project the seed investor finances; on the right, what the accelerator kept on the marginal project, which the seed investor stops financing.

We show that this term is strictly log-concave in ρ_a, so that the condition has a single solution. Its first factor, ρ_a, is log-concave. Its second factor is $G(\sigma_a + (1 - \rho_a)W(I)) - G(\sigma_a)$, the difference between G at a point affine in ρ_a and $G(\sigma_a)$. At a point $\ell > \sigma_a$, its log-derivative is

$$\frac{g(\ell)}{G(\ell) - G(\sigma_a)} = \Big[\int_0^{\ell - \sigma_a} \frac{g(\ell - s)}{g(\ell)}\, ds\Big]^{-1}.$$

When ℓ rises, the bound rises and each ratio $g(\ell - s)/g(\ell)$ rises, its log-derivative being positive since $(\log g)'$ decreases (Bagnoli and Bergstrom, 2005). The integral rises, and the log-derivative falls: the second factor is log-concave, and the product is strictly log-concave since the first factor is. The condition characterizes the unique maximum $\rho_a^*(\sigma_a)$.

(ii) An optimum exists: Π_a is continuous, $\Pi_a(0, \rho_a) \geq 0$, and for $\sigma_a \geq I$ the bound of the proof of Proposition 3 gives $\Pi_a \leq (1 - G(\sigma_a))(W(I) - \sigma_a) - G(\sigma_a)(I - E(\pi)) < 0$, since $W(I) < I$. The maximum is reached on the compact set $[0, I] \times [0, 1]$. At fixed

ρ_a, the derivative of Π_a in σ_a is

$$B'(\sigma_a) + \rho_a W(I)\Big[g\big(\sigma_a + (1-\rho_a)W(I)\big) - g(\sigma_a)\Big] \\ = (1-\rho_a)W(I)\, g(\sigma_a) + \rho_a W(I)\, g\big(\sigma_a + (1-\rho_a)W(I)\big) \\ - \big(1 - G(\sigma_a)\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)\, g(\ell)\, d\ell,$$

by the expression of B' in the proof of Proposition 3. At a positive optimum, it vanishes at $\rho_a = \rho_a^*$, by the envelope theorem.

(iii) At σ_a^B, any $\rho_a \in (0,1)$ adds $\rho_a W(I)[G(\sigma_a^B + (1-\rho_a)W(I)) - G(\sigma_a^B)] > 0$ to $B(\sigma_a^B)$. Hence $\Pi_a(\sigma_a^*, \rho_a^*) > B(\sigma_a^B)$. □

Proof of Proposition 7

We show first that, in the state $\pi > x$, the protection leaves

$$\Big(\pi - x - \frac{\sigma_a}{\lambda_a}\Big)^+ - \frac{\sigma_a}{V_a}\Big(\pi - x - \frac{V_a}{\lambda_a}\Big)^+.$$

Since $1/\min(a,b) = \max(1/a, 1/b)$, the claim times $\pi - x$ is $\max(\sigma_a(\pi-x)/V_a,\ \sigma_a/\lambda_a)$, and the protection takes this value, at most $\pi - x$. Since $V_a \geq \sigma_a$, $\sigma_a/\lambda_a \leq V_a/\lambda_a$, and there are three zones. If $\pi - x \leq \sigma_a/\lambda_a$, the protection takes everything. If $\sigma_a/\lambda_a \leq \pi - x \leq V_a/\lambda_a$, then $\sigma_a(\pi - x)/V_a \leq \sigma_a/\lambda_a$, and the protection leaves $\pi - x - \sigma_a/\lambda_a$. If $\pi - x \geq V_a/\lambda_a$, it leaves $(1 - \sigma_a/V_a)(\pi - x)$. The stated expression takes these three values: in the third zone, it equals $\pi - x - \sigma_a/\lambda_a - (\sigma_a/V_a)(\pi - x) + \sigma_a/\lambda_a$. Since $\int_x^1 (\pi - x - y)^+ f(\pi)\, d\pi = W(x+y)$ for $y \geq 0$, by the definition of W, the protection leaves $W(x + \sigma_a/\lambda_a) - (\sigma_a/V_a)W(x + V_a/\lambda_a)$ in expectation.

We turn to the seed investor, and bound first its profit. Its share is at most what the protection leaves, so its profit is at most $W(x + \sigma_a/\lambda_a) - (\sigma_a/V_a)W(x + V_a/\lambda_a) - \sigma_s$. Since $W' = -(1-F)$ and x falls by one dollar per dollar invested, this bound has derivative

$$-F\Big(x + \frac{\sigma_a}{\lambda_a}\Big) - \frac{\sigma_a}{V_a}\Big(1 - F\Big(x + \frac{V_a}{\lambda_a}\Big)\Big) \leq 0$$

in σ_s, and is maximal at the need, $\sigma_s = \ell - \sigma_a$ and $x = I$.

This bound is attained. A priced SAFE with $\rho_s = 1$ receives all the equity the protection leaves. An unpriced SAFE with cap $\ell - \sigma_a$ claims $(\ell - \sigma_a)/\min(\ell - \sigma_a, \lambda(\pi - I)) \geq 1$ in every state, and receives it too. Both attain the bound at the need, which is nonnegative if and only if the stated condition holds.

We turn to the accelerator. On each project the seed investor finances, the accelerator keeps the value the protection takes at the threshold I,

$$W(I) - W\Big(I + \frac{\sigma_a}{\lambda_a}\Big) + \frac{\sigma_a}{V_a} W\Big(I + \frac{V_a}{\lambda_a}\Big),$$

and the seed investor finances the shocks $\sigma_a < \ell \leq \sigma_a + W(I+\sigma_a/\lambda_a) - (\sigma_a/V_a)W(I+V_a/\lambda_a)$. Hence the accelerator's profit is the profit of Section 4 with $\rho_a W(I)$ replaced by this value: the two protections differ only by the values they allow at a given contribution.

We find next the values the SAFE allows. If $\sigma_a = 0$, the claim is zero, and the accelerator keeps nothing. Suppose $\sigma_a > 0$. We show that the value kept covers $(W(I) - W(I+\sigma_a), W(I)]$.

- The protection leaves at most $W(I+\sigma_a/\lambda_a) \leq W(I+\sigma_a)$, since W decreases and $\lambda_a \leq 1$: the value kept is at least $W(I) - W(I+\sigma_a)$. With $\lambda_a = 1$, it equals $W(I) - W(I+\sigma_a) + (\sigma_a/V_a)W(I+V_a)$, which tends to this bound as $V_a \to +\infty$.
- With $V_a = \sigma_a$, the value kept is $W(I)$.
- The value kept is continuous in (V_a, λ_a), on the connected set $V_a \geq \sigma_a$, $\lambda_a \in (0,1]$. The values it takes form an interval. This interval contains $W(I)$, and values as close as we want to $W(I) - W(I+\sigma_a)$, so it contains $(W(I) - W(I+\sigma_a), W(I)]$.

We compare last the SAFE with the fixed share. Let (σ_a^P, ρ_a^P) be an optimal fixed share, with profit Π_a^P. We show first that no SAFE earns more than Π_a^P. With a contribution $\sigma_a > 0$, the value the SAFE keeps lies in $[0, W(I)]$, and the fixed share equal to this value divided by $W(I)$ lies in $[0,1]$: with the same contribution, this fixed share keeps the same value and earns the same profit, which is at most Π_a^P. Without contribution, the SAFE keeps nothing and earns what the fixed share $\rho_a = 0$ earns without contribution, which is also at most Π_a^P.

We show next that, if $\sigma_a^P > 0$ and $\rho_a^P W(I) > W(I) - W(I+\sigma_a^P)$, a SAFE earns Π_a^P. The value $\rho_a^P W(I)$ is then above $W(I) - W(I+\sigma_a^P)$ and at most $W(I)$, so that a SAFE with contribution σ_a^P keeps it, and this SAFE earns Π_a^P since it has the contribution and the value kept of the optimal fixed share.

We show last that, if $\sigma_a^P = 0$, a SAFE comes arbitrarily close to Π_a^P. Without contribution, the fixed share ρ_a earns $\rho_a W(I)[G((1-\rho_a)W(I)) - G(0)]$, which is zero at $\rho_a = 0$ and positive for $\rho_a \in (0,1)$ since $g > 0$, so that $\rho_a^P > 0$. Since W is continuous, $W(I) - W(I+\sigma_a)$ tends to 0 with σ_a, and for $\sigma_a > 0$ small enough it is below $\rho_a^P W(I)$, so that a SAFE with contribution σ_a keeps exactly $\rho_a^P W(I)$ and earns $B(\sigma_a) + \rho_a^P W(I)[G(\sigma_a + (1-\rho_a^P)W(I)) - G(\sigma_a)]$. In $B(\sigma_a) = \int_0^{\sigma_a} W(I+\ell-\sigma_a)\, g(\ell)\, d\ell - \sigma_a$, the integrand is at most $W(0)$ times the maximum of g, which exists since g is log-concave, on an interval of length σ_a, so that $B(\sigma_a)$ tends to 0. Since G is continuous, the profit of the SAFE tends to $\rho_a^P W(I)[G((1-\rho_a^P)W(I)) - G(0)] = \Pi_a^P$. □

Proof of Proposition 8

ρ_a^0 and m are chosen by the accelerator before the round: here, they are parameters.

We recall first the rule. Each SAFE claims a share of the equity held before the equity round. If the claims sum to at most $1-\rho_a^0$, each receives its claim. Otherwise, each claim is multiplied by $1-\rho_a^0$ divided by the sum of the claims, so that the shares received sum to $1-\rho_a^0$. In both cases, each claim is multiplied by the minimum of 1 and of $1-\rho_a^0$ divided by the sum of the claims. The seed investor's share is therefore the minimum of its claim and of its claim times $1-\rho_a^0$ divided by the sum of the claims.

We compute next the shares. With an unpriced SAFE, the two SAFEs convert at the same price, $\min(V_s, \lambda(\pi-x))$. The seed investor claims $\sigma_s/\min(V_s, \lambda(\pi-x))$, and the claims sum to $(\sigma_s+m)/\min(V_s, \lambda(\pi-x))$. The seed investor's claim times $1-\rho_a^0$ divided by this sum is

$$\frac{\sigma_s}{\min(V_s, \lambda(\pi-x))} \cdot \frac{(1-\rho_a^0)\min(V_s, \lambda(\pi-x))}{\sigma_s+m} = \frac{(1-\rho_a^0)\,\sigma_s}{\sigma_s+m}.$$

By the rule, its share is

$$\min\Big(\frac{\sigma_s}{\min(V_s, \lambda(\pi-x))},\ \frac{(1-\rho_a^0)\,\sigma_s}{\sigma_s+m}\Big).$$

With a priced SAFE, the seed investor converts at its valuation σ_s/ρ_s and claims ρ_s. The MFN component converts at $\min(\sigma_s/\rho_s, \pi-x)$ and claims $m/\min(\sigma_s/\rho_s, \pi-x)$.

In the states $\pi-x \geq \sigma_s/\rho_s$, the MFN component claims $m\rho_s/\sigma_s$, and the claims sum to $\rho_s(\sigma_s+m)/\sigma_s$. The claim ρ_s times $1-\rho_a^0$ divided by this sum is $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$, and, by the rule, the seed investor's share is $\min\big(\rho_s, (1-\rho_a^0)\sigma_s/(\sigma_s+m)\big)$.

In the states $\pi-x < \sigma_s/\rho_s$, the MFN component claims $m/(\pi-x)$, and the claims sum to $\rho_s + m/(\pi-x)$. The claim ρ_s times $1-\rho_a^0$ divided by this sum is, multiplying the numerator and the denominator by $\pi-x$,

$$\frac{(1-\rho_a^0)\,\rho_s}{\rho_s + m/(\pi-x)} = \frac{(1-\rho_a^0)\,\rho_s(\pi-x)}{\rho_s(\pi-x)+m},$$

and, by the rule, the seed investor's share is

$$\min\Big(\rho_s,\ \frac{(1-\rho_a^0)\,\rho_s(\pi-x)}{\rho_s(\pi-x)+m}\Big).$$

We show that, when $m>0$, this share is strictly below $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$. The function $t \mapsto t/(t+m)$ has derivative $m/(t+m)^2 > 0$, so it is strictly increasing. In these states, $\rho_s(\pi-x) < \sigma_s$. Hence $\rho_s(\pi-x)/(\rho_s(\pi-x)+m) < \sigma_s/(\sigma_s+m)$, and, multiplying by $1-\rho_a^0$, the second term of the minimum is strictly below $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$. The share, at most this second term, is too.

Hence, in each of the three cases, the seed investor's share is a minimum whose second term is at most $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$.

We bound first the seed investor's profit. Under both contracts, the share never exceeds $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$. What the seed investor receives, its share times $\pi - x$ in the states $\pi > x$, therefore has an expectation of at most $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ times $W(x)$, and the profit at the amount σ_s is at most

$$\bar{\Pi}_s(\sigma_s) = \frac{(1-\rho_a^0)\,\sigma_s}{\sigma_s+m}\,W(x) - \sigma_s.$$

We compute next the derivative of this bound in σ_s. The cap $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ has derivative $(1-\rho_a^0)m/(\sigma_s+m)^2$. Since $x = I+\ell-\sigma_a-\sigma_s$ falls by one dollar per dollar invested and $W'(x) = -(1-F(x))$, $W(x)$ has derivative $1-F(x)$ in σ_s. The derivative of the bound is therefore

$$(1-\rho_a^0)\Big[\frac{m}{(\sigma_s+m)^2}W(x) + \frac{\sigma_s\big(1-F(x)\big)}{\sigma_s+m}\Big] - 1.$$

The first term is the gain in share: each dollar the seed investor adds dilutes the MFN component, and this gain is large when m weighs a lot against σ_s.

We show next that the bound does not rise beyond $x = 0$: its derivative is nonpositive when $x \le 0$, that is, when $\sigma_s \ge I+\ell-\sigma_a$. For $x \le 0$, $F(x) = 0$, and $W(x) = E(\pi) - x = E(\pi) - I - (\ell-\sigma_a) + \sigma_s < \sigma_s$ by (A1). The derivative is therefore at most

$$(1-\rho_a^0)\Big[\frac{m\,\sigma_s}{(\sigma_s+m)^2} + \frac{\sigma_s}{\sigma_s+m}\Big] - 1 = (1-\rho_a^0)\,\frac{\sigma_s(\sigma_s+2m)}{(\sigma_s+m)^2} - 1,$$

which is nonpositive, since $\sigma_s(\sigma_s+2m) = (\sigma_s+m)^2 - m^2$. The bound reaches its maximum on the compact interval $\ell-\sigma_a \le \sigma_s \le I+\ell-\sigma_a$.

We show next that the seed investor does not finance the large shocks. Suppose $\ell-\sigma_a+m > (1-\rho_a^0)W(I)$. Since $W' = -(1-F)$ and $0 \le 1-F \le 1$,

$$W(x) - W(I) = \int_x^I \big(1-F(t)\big)\,dt \le I - x = \sigma_s - (\ell-\sigma_a).$$

Multiplying by $1-\rho_a^0$, then using the assumption, $1-\rho_a^0 \le 1$ and $\sigma_s - (\ell-\sigma_a) \ge 0$,

$$\begin{aligned}(1-\rho_a^0)W(x) &\le (1-\rho_a^0)W(I) + (1-\rho_a^0)\big(\sigma_s-(\ell-\sigma_a)\big)\\ &< (\ell-\sigma_a) + m + \sigma_s - (\ell-\sigma_a) = \sigma_s + m.\end{aligned}$$

Multiplying by $\sigma_s/(\sigma_s+m)$, $(1-\rho_a^0)\sigma_s W(x)/(\sigma_s+m) < \sigma_s$: the bound is negative at every $\sigma_s \ge \ell-\sigma_a$, and the seed investor loses money at every amount, with either contract. If $\ell-\sigma_a+m \le (1-\rho_a^0)W(I)$, the bound at the need, $(\ell-\sigma_a)[(1-\rho_a^0)W(I)/(\ell-\sigma_a+m) - 1]$, is nonnegative.

Last, if the derivative of the bound is positive at $\sigma_s = \ell-\sigma_a$, where $x = I$, the bound rises beyond the need, and its maximum exceeds $\ell-\sigma_a$. This derivative is the left side of the condition of the proposition.

We show next that the unpriced SAFE attains the bound. Fix σ_s, take $\lambda = 1$ and $V_s = (\sigma_s + m)/(1 - \rho_a^0)$. In every state, $\min(V_s, \pi - x) \leq V_s$, so $\sigma_s / \min(V_s, \pi - x) \geq \sigma_s / V_s = (1 - \rho_a^0)\sigma_s/(\sigma_s + m)$: the minimum giving the share equals its second term, and the share equals $(1 - \rho_a^0)\sigma_s/(\sigma_s + m)$ everywhere. By the shares above, no (V_s, λ) gives a share above this cap: a discount brings the seed investor nothing. With an unpriced SAFE, the seed investor's profit at the amount σ_s is therefore $\bar{\Pi}_s(\sigma_s)$, and it finances the shock if and only if $\ell - \sigma_a + m \leq (1 - \rho_a^0)W(I)$.

We show last that the priced SAFE does strictly worse when $m > 0$. With $m = 0$, the clause is a fixed share, and both contracts give the same profit (Propositions 4 and 5). Suppose $m > 0$. The priced SAFE's profit is continuous in (σ_s, ρ_s) on a compact set, and has a maximum, at most the maximum of $\bar{\Pi}_s$. Suppose that they are equal, at the priced SAFE's optimum σ_s^p. The states $x < \pi < x + \sigma_s^p/\rho_s$ have positive probability, and in them the priced SAFE's share is strictly below $(1 - \rho_a^0)\sigma_s^p/(\sigma_s^p + m)$: its profit is strictly below $\bar{\Pi}_s(\sigma_s^p)$, which contradicts the equality. □

Proof of Proposition 9

By Proposition 8, the seed investor uses an unpriced SAFE.

By Proposition 8, the seed investor finances exactly the shocks $\sigma_a < \ell \leq \sigma_a + (1 - \rho_a^0)W(I) - m$, where it may invest $\sigma_s^* > \ell - \sigma_a$ and takes $\rho_s^* = (1 - \rho_a^0)\sigma_s^*/(\sigma_s^* + m)$. On these projects, the accelerator keeps $(1 - \rho_s^*)W(x^*)$, where ρ_s^* and $x^* = I + \ell - \sigma_a - \sigma_s^*$ depend only on $\ell - \sigma_a$, ρ_a^0 and m. Hence

$$\Pi_a(\sigma_a, \rho_a^0, m) = B(\sigma_a) + \int_{\sigma_a}^{\sigma_a + (1-\rho_a^0)W(I) - m} (1 - \rho_s^*)W(x^*)\, g(\ell)\, d\ell.$$

For $m = 0$, this is the program of Proposition 6.

On the last financed project, $\ell - \sigma_a + m = (1 - \rho_a^0)W(I)$. We show that the seed investor invests exactly its need. The inequality used for the shocks it does not finance, with this equality instead of the strict assumption, gives

$$(1 - \rho_a^0)W(x) \leq (\ell - \sigma_a) + m + (1 - \rho_a^0)\big(\sigma_s - (\ell - \sigma_a)\big) \leq \sigma_s + m,$$

and, multiplying by $\sigma_s/(\sigma_s + m)$, the bound $\bar{\Pi}_s(\sigma_s)$ of the proof of Proposition 8 is nonpositive at every amount and zero at the need. Beyond the need, $x < I$, and the first inequality is strict: $F(I) > 0$ by (A1), F is continuous, hence positive on an interval to the left of I, and $W(x) - W(I) = (I - x) - \int_x^I F(t)\, dt < I - x$. The bound is strictly negative there. Since the unpriced SAFE attains it, the seed investor invests its need and earns zero, and the accelerator keeps $W(I) - (\ell - \sigma_a) = \rho_a^0 W(I) + m$. The last financed project moves back by $W(I)$ per unit of ρ_a^0, and by one unit per unit of m. Lowering ρ_a^0 by one unit and raising m by $W(I)$ leaves it in place.

We show next that the MFN component lowers the value kept on no financed project. The seed investor invests at least $\ell - \sigma_a$, so $x^* \leq I$ and $W(x^*) \geq W(I)$. Its share is

at most $(1-\rho_a^0)\sigma_s^*/(\sigma_s^*+m) \le 1-\rho_a^0$. Hence the accelerator keeps at least $\rho_a^0 W(I)$, what it keeps with $m=0$.

The improvement. Take a fixed share (σ_a, ρ_a) with $\sigma_a > 0$ and $0 < \rho_a < 1$. We show that a small MFN component strictly raises the profit. Keep σ_a and $\rho_a^0 = \rho_a$, and add $m \in (0, \sigma_a)$. With $m=0$, the seed investor invests its need on the rounds $0 < \ell - \sigma_a \le (1-\rho_a^0)W(I)$, and the accelerator keeps $\rho_a^0 W(I)$ on each (Proposition 4). The seed investor's zone becomes $0 < \ell - \sigma_a \le (1-\rho_a^0)W(I) - m$. Fix a round $\delta \in (0, (1-\rho_a^0)W(I))$ and take $m < (1-\rho_a^0)W(I) - \delta$, so that the round δ lies in the zone with and without the MFN component. It separates the small rounds, $\ell - \sigma_a < \delta$, on which the seed investor may invest beyond its need, from the others, on which we show that it invests its need. The round δ does not depend on m: we first let m tend to 0 at a given δ.

- The projects with $\ell - \sigma_a \ge \delta$ and $\ell - \sigma_a + m \le (1-\rho_a^0)W(I)$ are financed. We show that, for small m, the seed investor invests exactly its need on these projects, using $\rho_a^0 > 0$.

  We show that large excess amounts lose. At $\sigma_s \ge \ell - \sigma_a$, the bound is at most $(1-\rho_a^0)W(x) - \sigma_s$, since $\sigma_s/(\sigma_s+m) \le 1$. Since $W(x) - W(I) = \int_x^I (1-F(t))\,dt$ and $I - x = \sigma_s - (\ell - \sigma_a)$,

  $$(1-\rho_a^0)W(x) - \sigma_s = (1-\rho_a^0)W(I) - (\ell - \sigma_a) - \int_x^I \big[1 - (1-\rho_a^0)(1-F(t))\big]\,dt,$$

  and the integrand is at least ρ_a^0. At the need, the bound equals $(1-\rho_a^0)W(I) - (\ell - \sigma_a) - (1-\rho_a^0)mW(I)/(\ell - \sigma_a + m)$. The bound at σ_s is therefore below the bound at the need as soon as $\sigma_s - (\ell - \sigma_a) > (1-\rho_a^0)mW(I)/(\rho_a^0\delta)$, a quantity that tends to zero with m.

  We show that small excess amounts lose too. When $\sigma_s - (\ell - \sigma_a)$ is at most this quantity, x lies within it below I, so $x \ge 0$ for small m and $W(x) \le W(0)$. The derivative of the bound is then at most $(1-\rho_a^0)[mW(0)/\delta^2 + 1 - F(x)] - 1$, which tends to $(1-\rho_a^0)(1-F(I)) - 1 < 0$ as $m \to 0$, uniformly in $\ell - \sigma_a \ge \delta$, since F is continuous. The bound decreases on this interval.

  Hence $\sigma_s^* = \ell - \sigma_a$ for small m, and the value kept gains

  $$\Big(1 - \frac{(1-\rho_a^0)(\ell - \sigma_a)}{\ell - \sigma_a + m}\Big)W(I) - \rho_a^0 W(I) = \frac{(1-\rho_a^0)\,m}{\ell - \sigma_a + m} W(I).$$

- On the projects with $\ell - \sigma_a < \delta$, the seed investor may invest beyond its need, but the value kept does not fall, since the MFN component lowers the value kept on no financed project.

- The projects financed for $m=0$ and no longer for $m>0$ lie in a band of width m, and the value kept on each is at most $W(I)$.

We show that the gain exceeds the loss, both divided by m.

The loss. Since g is log-concave, it is bounded, and the loss is at most $m\,W(I)\sup g$. Divided by m, it stays below $W(I)\sup g$, whatever δ and m.

The gain. On each round $\ell - \sigma_a \geq \delta$ of the zone, the accelerator gains $(1-\rho_a^0)\,m\,W(I)/(\ell-\sigma_a+m)$. The gain on these rounds, divided by m, is therefore

$$(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)-m}\frac{g(\ell)}{\ell-\sigma_a+m}\,d\ell.$$

As $m \to 0$, the upper bound tends to $\sigma_a + (1-\rho_a^0)W(I)$, and the integrand tends to $g(\ell)/(\ell-\sigma_a)$ while staying below $\sup g/\delta$, since $\ell - \sigma_a \geq \delta$. The gain divided by m therefore tends to

$$(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)}\frac{g(\ell)}{\ell-\sigma_a}\,d\ell.$$

This limit grows without bound as δ falls: near $\ell = \sigma_a$, the integrand is close to $g(\sigma_a)/(\ell-\sigma_a)$, with $g(\sigma_a) > 0$, whose integral diverges.

The choice of δ, then of m. Take δ such that this limit exceeds $W(I)\sup g$. At this δ, for m small enough, the gain divided by m still exceeds it, hence exceeds the loss divided by m, and the profit strictly rises.

(i) Apply the improvement to the optimal fixed share (σ_a^P, ρ_a^P) of Proposition 6, with $\sigma_a^P > 0$ by assumption and $0 < \rho_a^P < 1$ by its point (i): an MFN clause earns strictly more than the optimal fixed share, which rules out $m^* = 0$. The accelerator earns strictly more with an MFN clause than with any fixed share, hence than in the benchmark by Proposition 6. The argument does not assume that an optimum with an MFN clause exists.

We compute next how the value kept reacts to ρ_a^0 and m. On a financed project, the accelerator keeps the project's value minus the seed investor's amount and profit:

$$(1-\rho_s^*)W(x^*) = W(x^*) - \sigma_s^* - \Pi_s(\sigma_s^*).$$

The first two terms have derivative $-F(x^*)$ in σ_s. The value kept therefore has derivative in ρ_a^0

$$-\frac{\partial\Pi_s}{\partial\rho_a^0}(\sigma_s^*) - \Big[F(x^*) + \frac{\partial\Pi_s}{\partial\sigma_s}(\sigma_s^*)\Big]\frac{\partial\sigma_s^*}{\partial\rho_a^0},$$

and the same expression in m. Two kinds of shocks arise.

- Without overinvestment, $\sigma_s^* = \ell - \sigma_a$ does not move, and only the first term remains.
- With interior overinvestment, $\partial\Pi_s/\partial\sigma_s = 0$. Where $\partial^2\Pi_s/\partial\sigma_s^2$ does not vanish,

the implicit function theorem gives, at $\sigma_s = \sigma_s^*$ and $x = x^*$,

$$\frac{\partial \sigma_s^*}{\partial \rho_a^0} = \frac{1}{(1-\rho_a^0)\,\partial^2 \Pi_s / \partial \sigma_s^2} < 0,$$
$$\frac{\partial \sigma_s^*}{\partial m} = -\frac{(1-\rho_a^0)\big[(\sigma_s - m)W(x) - \sigma_s(\sigma_s + m)(1-F(x))\big]}{(\sigma_s + m)^3\,\partial^2 \Pi_s / \partial \sigma_s^2},$$

since $\partial^2 \Pi_s / \partial \sigma_s \partial \rho_a^0 = -(\partial \Pi_s / \partial \sigma_s + 1)/(1-\rho_a^0) = -1/(1-\rho_a^0)$ at the critical point.

With

$$\frac{\partial \Pi_s}{\partial \rho_a^0} = -\frac{\sigma_s W(x)}{\sigma_s + m}, \qquad \frac{\partial \Pi_s}{\partial m} = -\frac{(1-\rho_a^0)\,\sigma_s W(x)}{(\sigma_s + m)^2},$$

a stronger protection always raises the value kept: $\partial \Pi_s / \partial \rho_a^0 < 0$, $\partial \sigma_s^* / \partial \rho_a^0 \leq 0$, and $F(x^*) + \partial \Pi_s / \partial \sigma_s \geq 0$. It also curbs overinvestment.

(ii) As $\ell - \sigma_a \to 0$, the left side of the overinvestment condition tends to $(1 - \rho_a^0)W(I)/m$, which exceeds 1 since the zone is not empty: $m < (1-\rho_a^0)W(I)$. With $m^* > 0$, there are therefore always shocks on which the seed investor invests beyond its need.

(iii) The last financed project, $\bar{\ell} = \sigma_a + (1-\rho_a^0)W(I) - m$, leaves the zone when ρ_a^0 or m rises, by $W(I)$ per unit of ρ_a^0 and by one unit per unit of m, and the accelerator kept $\rho_a^0 W(I) + m$ on it. With the integrals running over the financed shocks, the derivatives of the profit are

$$\text{(a)} \quad \frac{\partial \Pi_a}{\partial \rho_a^0} = \int \frac{\partial}{\partial \rho_a^0}\big[(1-\rho_s^*)W(x^*)\big]g(\ell)\,d\ell - \big[\rho_a^0 W(I) + m\big]\,g(\bar{\ell})\,W(I),$$
$$\text{(b)} \quad \frac{\partial \Pi_a}{\partial m} = \int \frac{\partial}{\partial m}\big[(1-\rho_s^*)W(x^*)\big]g(\ell)\,d\ell - \big[\rho_a^0 W(I) + m\big]\,g(\bar{\ell}).$$

Combining (a) and (b), the boundary term disappears:

$$\text{(c)} \quad W(I)\frac{\partial \Pi_a}{\partial m} - \frac{\partial \Pi_a}{\partial \rho_a^0} = \int \Big[W(I)\frac{\partial}{\partial m} - \frac{\partial}{\partial \rho_a^0}\Big]\big[(1-\rho_s^*)W(x^*)\big]g(\ell)\,d\ell.$$

At the optimum, with $\rho_a^{0*} > 0$, the derivative (a) vanishes. If $m^* < \sigma_a^*$, m can move in both directions and (b) vanishes, so (c) is zero. If $m^* = \sigma_a^*$, only a lower m is feasible at the given contribution, so (b) is nonnegative, and (c) is nonnegative.

(c) concerns the direction that lowers ρ_a^0 and raises m while leaving the last financed project in place. On a financed shock, its integrand equals

$$\frac{\sigma_s^* W(x^*)}{(\sigma_s^* + m)^2}\Big[(1-\rho_a^0)W(I) - m - \sigma_s^*\Big] - \Big[F(x^*) + \frac{\partial \Pi_s}{\partial \sigma_s}\Big]\Big[W(I)\frac{\partial \sigma_s^*}{\partial m} - \frac{\partial \sigma_s^*}{\partial \rho_a^0}\Big].$$

The first term is the gain from discrimination: at a given zone, the MFN component takes the more, the further the round is from the last financed one, at $\ell - \sigma_a =$

$(1-\rho_a^0)W(I)-m$. On a shock without overinvestment, $\sigma_s^* = \ell - \sigma_a$ is below this edge, the first term is strictly positive and the second is zero. The second term is the cost on the shocks with overinvestment, where the move changes the seed investor's amount, and each extra dollar is lost on the projects that are not fully funded. □

Proof of Proposition 10

Fix $\ell > \sigma_a$ and a seed round with amount $\sigma_s \geq \ell - \sigma_a$, threshold $x = I + \ell - \sigma_a - \sigma_s$, and any contract and protection. The seed investor receives at most the equity, worth $W(x)$ in expectation, so its profit is at most $W(x) - \sigma_s$. This bound has derivative $-F(x) \leq 0$ in σ_s: it is largest at $\sigma_s = \ell - \sigma_a$, where $x = I$, and equals $W(I) - (\ell - \sigma_a)$ there. The seed investor earns a nonnegative profit only if $\ell - \sigma_a \leq W(I)$. By (A1), $W(I) \leq E(\pi) < I$. □

Proof of Proposition 11

With an MFN clause against an unpriced SAFE, the seed investor finances exactly the shocks with $\ell - \sigma_a + m \leq (1-\rho_a^0)W(I)$ (Proposition 8), and the accelerator keeps $\rho_a^0 W(I) + m$ on the last one (proof of Proposition 9). With $m = 0$, this is the zone and the share of the fixed share $\rho_a = \rho_a^0$ (Proposition 4); with $\rho_a^0 = m = 0$, the zone of Proposition 1, where the accelerator keeps nothing. In all cases, the accelerator covers $[0, \sigma_a]$ alone, and the experiment is covered up to $\sigma_a + (1-\rho_a^0)W(I) - m$. □

Proof of Proposition 12

(i) When the seed investor covers only its need, its threshold is I (Propositions 1 and 4). If it finances the shocks up to $\bar{\ell}$, the probability of full funding is

$$\int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big) g(\ell)\, d\ell + \big(1 - F(I)\big)\big[G(\bar{\ell}) - G(\sigma_a)\big].$$

Its derivative in $\bar{\ell}$ is $(1 - F(I))\, g(\bar{\ell}) > 0$, since $F(I) < 1$, as $W(I) > 0$. Its derivative in σ_a, at a given $\bar{\ell}$, is $\int_0^{\sigma_a} f(I + \ell - \sigma_a)\, g(\ell)\, d\ell$, the terms $(1 - F(I))g(\sigma_a)$ of the two integrals canceling. With a fixed share ρ_a, the last shock is $\sigma_a + (1-\rho_a)W(I)$, and moves one for one with σ_a: at a given ρ_a, the derivative in σ_a is

$$\int_0^{\sigma_a} f(I + \ell - \sigma_a)\, g(\ell)\, d\ell + \big(1 - F(I)\big) g\big(\sigma_a + (1-\rho_a)W(I)\big) > 0.$$

Without protection, $\rho_a = 0$. We go from $(\sigma_a^B, 0)$ to (σ_a^P, ρ_a^P) in two steps. Raising the contribution from σ_a^B to σ_a^P at $\rho_a = 0$ gives the integral of the derivative above, the first term. Raising the share from 0 to ρ_a^P at σ_a^P moves the last shock from $\sigma_a^P + W(I)$ down to $\sigma_a^P + (1-\rho_a^P)W(I)$, and gives the second term, strictly negative since $\rho_a^P > 0$.

(ii) On a round where the seed investor invests $\sigma_s^* > \ell - \sigma_a$, the threshold is $x^* = I + \ell - \sigma_a - \sigma_s^* < I$. Since $f > 0$, $F(x^*) < F(I)$, and the project is fully funded with probability $1 - F(x^*) > 1 - F(I)$. □

Proof of Proposition 13

We write first the two probabilities. By Proposition 11, the probability of a seed round is $G(\sigma_a + (1-\rho_a^0)W(I) - m) - G(\sigma_a)$. Under the MFN clause, the probability of full funding is

$$\int_0^{\sigma_a^M} \big(1-F(I+\ell-\sigma_a^M)\big)g(\ell)\,d\ell + \big(1-F(I)\big)\Big[G\big(\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M\big)-G(\sigma_a^M)\Big] + \int_{\sigma_a^M}^{\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M} \big(F(I)-F(x^*(\ell))\big)g(\ell)\,d\ell,$$

and under the fixed share, where the seed investor invests its need (Proposition 4), the same expression with $(\sigma_a^P, \rho_a^P, 0)$ and without the last integral.

For the probability of a seed round, we add and subtract $G(\sigma_a^M + (1-\rho_a^P)W(I)) - G(\sigma_a^M)$ in the difference, which gives the stated sum.

For the probability of full funding, we go from the fixed share to the MFN clause in two steps. Raising the contribution from σ_a^P to σ_a^M at the share ρ_a^P gives the integral of the derivative in the proof of Proposition 12, the first term. Moving the last shock from $\sigma_a^M + (1-\rho_a^P)W(I)$ to $\sigma_a^M + (1-\rho_a^{0M})W(I) - m^M$ gives the second. The last integral is nonnegative, since $x^*(\ell) \le I$ on every round. □

Proof of Proposition 14

Write $\bar{\ell} = \sigma_a + (1-\rho_a^0)W(I) - m$ for the last shock the seed investor finances.

(i) On a shock ℓ, the seed investor invests σ_s^*, the threshold is $x^* = I + \ell - \sigma_a - \sigma_s^*$, and the accelerator keeps $(1-\rho_s^*)W(x^*) = W(x^*) - \sigma_s^* - \Pi_s^*(\ell)$. Since $W' = -(1-F)$ and $I - x^* = \sigma_s^* - (\ell - \sigma_a)$,

$$W(x^*) - \sigma_s^* = W(I) + \int_{x^*}^{I} \big(1-F(x)\big)dx - \sigma_s^* = W(I) - (\ell - \sigma_a) - \int_{x^*}^{I} F(x)\,dx.$$

By Proposition 11, the seed investor finances exactly $(\sigma_a, \bar{\ell}]$, and the accelerator covers $[0, \sigma_a]$ alone; adding $B(\sigma_a)$ gives the profit.

(ii) At $\bar{\ell}$, the seed investor earns zero and invests its need (proof of Proposition 9), so the integrand of (i) equals $W(I) - (\bar{\ell} - \sigma_a) = \rho_a^0 W(I) + m$ there, and $\bar{\ell}$ moves by

$-W(I)$ per unit of ρ_a^0. Differentiating (i),

$$\frac{\partial \Pi_a}{\partial \rho_a^0} = -\int_{\sigma_a}^{\bar{\ell}} \frac{\partial}{\partial \rho_a^0}\Big(\Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(x)\,dx\Big) g(\ell)\,d\ell - W(I)\big(\rho_a^0 W(I) + m\big) g(\bar{\ell}).$$

The optimal fixed share ρ_a at σ_a with the same last shock keeps $\rho_a W(I) = W(I) - (\bar{\ell} - \sigma_a) = \rho_a^0 W(I) + m$ on it, and satisfies $G(\bar{\ell}) - G(\sigma_a) = \rho_a W(I)\, g(\bar{\ell})$ (Proposition 6). Hence $W(I)(\rho_a^0 W(I) + m)\, g(\bar{\ell}) = \int_{\sigma_a}^{\bar{\ell}} W(I)\, g(\ell)\,d\ell$, and

$$\frac{\partial \Pi_a}{\partial \rho_a^0} = -\int_{\sigma_a}^{\bar{\ell}} \Big[W(I) + \frac{\partial}{\partial \rho_a^0}\Big(\Pi_s^*(\ell) + \int_{x^*(\ell)}^{I} F(x)\,dx\Big)\Big] g(\ell)\,d\ell.$$

Lowering ρ_a^0 extends the zone, and pays if and only if this derivative is negative. On a round without investment beyond the need, $x^* = I$, the lost surplus is zero, and $\partial \Pi_s^* / \partial \rho_a^0 = -(\ell - \sigma_a) W(I)/(\ell - \sigma_a + m)$; the integrand equals $mW(I)/(\ell - \sigma_a + m)$. On a round with investment beyond the need, the lost surplus has derivative $F(x^*)\,\partial \sigma_s^* / \partial \rho_a^0$, since x^* falls one for one with σ_s^*, and $\partial \sigma_s^* / \partial \rho_a^0 < 0$ by the proof of Proposition 9.

(iii) We differentiate (i) in σ_a, with ρ_a^0 and m fixed. The bracket of (i) is $W(I) - (\ell - \sigma_a) - \Pi_s^*(\ell) - \int_{x^*(\ell)}^{I} F(x)\,dx$. The seed investor sees the shock only through the round, so its profit Π_s^* and its threshold x^* are functions of the round $t = \ell - \sigma_a$. The width of the zone, $(1-\rho_a^0)W(I) - m$, does not depend on σ_a. Changing variable to the round $t = \ell - \sigma_a$, the integral of (i) is $\int_0^{(1-\rho_a^0)W(I)-m} [W(I) - t - \Pi_s^*(t) - \int_{x^*(t)}^{I} F(x)\,dx] g(\sigma_a + t)\,dt$, with derivative

$$\int_0^{(1-\rho_a^0)W(I)-m} \Big(W(I) - t - \Pi_s^*(t) - \int_{x^*(t)}^{I} F(x)\,dx\Big) g'(\sigma_a + t)\,dt.$$

An integration by parts gives

$$\int_0^{(1-\rho_a^0)W(I)-m} (W(I) - t)\, g'(\sigma_a + t)\,dt = \big(\rho_a^0 W(I) + m\big) g(\bar{\ell}) - W(I)\, g(\sigma_a) + G(\bar{\ell}) - G(\sigma_a).$$

Adding $B'(\sigma_a) = W(I) g(\sigma_a) - (1 - G(\sigma_a)) - \int_0^{\sigma_a} F(I + \ell - \sigma_a) g(\ell)\,d\ell$ gives the stated derivative.

With a fixed share, $x^* = I$ and $\Pi_s^*(t) = (1 - \rho_a) W(I) - t$. An integration by parts gives

$$-\int_0^{(1-\rho_a)W(I)} \big((1 - \rho_a) W(I) - t\big) g'(\sigma_a + t)\,dt = (1 - \rho_a) W(I) g(\sigma_a) - \big[G(\bar{\ell}) - G(\sigma_a)\big],$$

and the derivative reduces to the one in the proof of Proposition 6. □

Proof of Proposition 15

We show that the total surplus of each covered project is $W(I) - \ell - \int_x^I F(t)\,dt$, then sum over the shocks.

We compute first the total surplus of a covered project. On a covered project with threshold x, the accelerator and the seed investor bring $I + \ell - x$ before the equity round, and the equity investor brings x for an equity worth x in expectation. The payoffs of the three agents add up to $W(x) - (I + \ell - x)$. Since $W' = -(1 - F)$,

$$W(x) = W(I) + \int_x^I \big(1 - F(t)\big)\,dt = W(I) + (I - x) - \int_x^I F(t)\,dt,$$

and the total surplus of the project is $W(I) - \ell - \int_x^I F(t)\,dt$. An uncovered project is abandoned, and its surplus is $-\sigma_a$, the contribution spent.

We sum next over the shocks. The accelerator covers $[0, \sigma_a]$ alone at the threshold $I+\ell-\sigma_a$, and the seed investor finances $(\sigma_a, \bar{\ell}]$ at the threshold $x^*(\ell)$ (Proposition 11). Summing,

$$TS = \int_0^{\bar{\ell}} \big(W(I) - \ell\big)\, g(\ell)\,d\ell - L_{acc} - L_{over} - \sigma_a\big(1 - G(\bar{\ell})\big),$$

and the first integral is $TS^* - L_{cov}$. If $\bar{\ell} < W(I)$, L_{cov} integrates the positive $W(I) - \ell$ over $(\bar{\ell}, W(I))$; if $\bar{\ell} > W(I)$, it equals $\int_{W(I)}^{\bar{\ell}} (\ell - W(I)) g(\ell)\,d\ell \geq 0$.

We compute last the derivatives. When the seed investor invests its need, $L_{over} = 0$, and the total surplus depends on the contribution and on the last shock:

$$\int_0^{\bar{\ell}} \big(W(I) - \ell\big)\, g(\ell)\,d\ell - L_{acc}(\sigma_a) - \sigma_a\big(1 - G(\bar{\ell})\big).$$

Its derivative in $\bar{\ell}$ is $(W(I) - \bar{\ell} + \sigma_a)\, g(\bar{\ell})$: the value of the last project at the threshold I net of its round, against $-\sigma_a$ if it were abandoned. With $\bar{\ell} = \sigma_a + W(I) - (\rho_a^0 W(I) + m)$, a dollar more kept on the last project lowers $\bar{\ell}$ by one dollar, and changes the total surplus by $-(\rho_a^0 W(I) + m)\, g(\bar{\ell})$. A dollar more of contribution, at a given amount kept, raises $\bar{\ell}$ by one dollar: the total surplus changes by $(\rho_a^0 W(I) + m) g(\bar{\ell})$ through the last shock, by $-(1 - G(\bar{\ell}))$ through the last term, and by $-dL_{acc}/d\sigma_a = -\int_0^{\sigma_a} F(I + \ell - \sigma_a) g(\ell)\,d\ell$ through L_{acc}, by Leibniz, the boundary term vanishing at $\ell = \sigma_a$. □

Proof of Proposition 16

Without protection and with the fixed share, the seed investor invests its need (Propositions 1 and 4). We go from $(\sigma_a^B, 0)$ to (σ_a^P, ρ_a^P) in two steps. Raising the contribution from σ_a^B to σ_a^P without protection, where the accelerator keeps nothing on the last

project, gives the integral of $-(1-G(s+W(I)))-\int_0^s F(I+\ell-s)g(\ell)\,d\ell$, the first term. Raising the share from 0 to ρ_a^P at σ_a^P moves the last shock from $\sigma_a^P+W(I)$ down to $\sigma_a^P+(1-\rho_a^P)W(I)$, and loses on each shock in between its value $\sigma_a^P+W(I)-\ell$ against abandonment, the second term, strictly negative since $\rho_a^P>0$. □

Proof of Proposition 17

With the MFN clause, the total surplus is the expression of the proof of Proposition 15 minus L_{over}, and with the fixed share the same expression with $L_{over}=0$. We go from the fixed share to the MFN clause in two steps. Raising the contribution from σ_a^P to σ_a^M at the share ρ_a^P gives the integral of the derivative in the contribution of the proof of Proposition 15, with $\rho_a^0 W(I)+m=\rho_a^P W(I)$, the first term. Moving the last shock from $\sigma_a^M+(1-\rho_a^P)W(I)$ to $\sigma_a^M+(1-\rho_a^{0M})W(I)-m^M$ adds or removes, on each shock in between, its value $\sigma_a^M+W(I)-\ell$ against abandonment, the second term. L_{over} is nonnegative since $x^*(\ell)\le I$ on every round. □

Proof of Proposition 18

Against a priced SAFE of valuation σ_s/ρ_s, the seed investor's share is at most $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ (proof of Proposition 8). On a financed project, with the seed investor's amount σ_s^p, threshold x^p and profit Π_s^p, the accelerator keeps

$$W(I)-(\ell-\sigma_a)-\Pi_s^p-\int_{x^p}^I F(t)\,dt,$$

since $W(x^p)=W(I)+\sigma_s^p-(\ell-\sigma_a)-\int_{x^p}^I F(t)\,dt$. Its share never falls below ρ_a^0, so it keeps at least $\rho_a^0 W(I)$, what it keeps with $m=0$.

Take a fixed share (σ_a,ρ_a^0) with $\sigma_a>0$ and $0<\rho_a^0<1$. The seed investor finances the rounds $0<\ell-\sigma_a\le(1-\rho_a^0)W(I)$, where the accelerator keeps $\rho_a^0W(I)$. We add a small $m\in(0,\sigma_a)$ and show that the profit strictly rises. As in the proof of Proposition 9, fix a round $\delta\in(0,(1-\rho_a^0)W(I))$, which does not depend on m; it separates the small rounds, $\ell-\sigma_a<\delta$, on which we only show that the value kept does not fall, from the others, on which we compute its gain.

We show first that the seed investor still finances almost all of the zone. On the round $\ell-\sigma_a$, consider the contract that invests the need for the share $\rho_s=(1-\rho_a^0)(\ell-\sigma_a)/(\ell-\sigma_a+m)$, of valuation $(\ell-\sigma_a+m)/(1-\rho_a^0)$. Above this valuation, the seed investor takes ρ_s. Below, it takes $(1-\rho_a^0)\rho_s(\pi-I)/(\rho_s(\pi-I)+m)$, and, using $1-\rho_a^0-\rho_s=(1-\rho_a^0)m/(\ell-\sigma_a+m)$, it misses

$$\frac{\rho_s\,m}{\rho_s(\pi-I)+m}\Big[1-\frac{(1-\rho_a^0)(\pi-I)}{\ell-\sigma_a+m}\Big],$$

which, multiplied by $\pi - I$, is at most m. Its profit with this contract is at least

$$\frac{(1-\rho_a^0)W(I)\,(\ell-\sigma_a)}{\ell-\sigma_a+m} - (\ell-\sigma_a) - m,$$

which is nonnegative if and only if $\sqrt{(1-\rho_a^0)W(I)(\ell-\sigma_a)} \geq \ell-\sigma_a+m$. The difference of the two sides is concave in $\ell-\sigma_a$; for $m \leq (1-\rho_a^0)W(I)/4$, it is nonnegative at $\ell-\sigma_a = 4m^2/((1-\rho_a^0)W(I))$ and at $\ell-\sigma_a = (1-\rho_a^0)W(I)-3m$, hence between them. The priced seed investor, which does at least as well, finances every round in between.

We show next that, on the rounds $\ell-\sigma_a \geq \delta$, the value kept gains at least $(1-\rho_a^0)\,m\,W(I)/(\ell-\sigma_a+m)$. The seed investor's share is at most $(1-\rho_a^0)\sigma_s^p/(\sigma_s^p+m)$, so Π_s^p is at most this share times $W(x^p)$, minus σ_s^p. With the expression of $W(x^p)$ above and $\int_{x^p}^I F(t)\,dt \leq F(I)(\sigma_s^p-(\ell-\sigma_a))$,

$$\Pi_s^p + \int_{x^p}^I F(t)\,dt \leq \frac{(1-\rho_a^0)\sigma_s^p}{\sigma_s^p+m}\big[W(I)+\sigma_s^p-(\ell-\sigma_a)\big] - \sigma_s^p \\ + \Big(1-\frac{(1-\rho_a^0)\sigma_s^p}{\sigma_s^p+m}\Big)F(I)\big(\sigma_s^p-(\ell-\sigma_a)\big).$$

Moreover $x^p \geq 0$: otherwise $W(x^p) = E(\pi)-x^p$ and, since $E(\pi) < I$, the profit would be negative. Hence $\sigma_s^p-(\ell-\sigma_a) \leq I$. The upper bound has derivative in σ_s^p

$$\frac{(1-\rho_a^0)\,m}{(\sigma_s^p+m)^2}\Big[W(I)+\big(1-F(I)\big)\big(\sigma_s^p-(\ell-\sigma_a)\big)\Big] - \Big(1-\frac{(1-\rho_a^0)\sigma_s^p}{\sigma_s^p+m}\Big)\big(1-F(I)\big),$$

which is negative on the rounds $\ell-\sigma_a \geq \delta$ as soon as $m < \rho_a^0\big(1-F(I)\big)\delta^2/\big((1-\rho_a^0)(W(I)+I)\big)$, since $\sigma_s^p \geq \delta$ and the last factor in parentheses is at least $\rho_a^0 > 0$. The bound is then maximal at $\sigma_s^p = \ell-\sigma_a$, and the value kept is at least

$$\rho_a^0 W(I) + \frac{(1-\rho_a^0)\,m\,W(I)}{\ell-\sigma_a+m}.$$

We show last that the gain exceeds the loss, both divided by m.

The small rounds. On the financed rounds with $\ell-\sigma_a < \delta$, the accelerator's share never falls below ρ_a^0: the value kept does not fall.

The loss. The rounds lost lie in two bands, of widths $4m^2/((1-\rho_a^0)W(I))$ and $3m$, at most $4m$ in total for $m \leq (1-\rho_a^0)W(I)/4$. The accelerator kept at most $W(I)$ on each, and g, log-concave, is bounded: the loss divided by m stays below $4W(I)\sup g$, whatever δ and m.

The gain. On each financed round $\ell-\sigma_a \geq \delta$, the value kept gains at least $(1-\rho_a^0)\,m\,W(I)/(\ell-\sigma_a+m)$. The gain on these rounds, divided by m, is therefore at least

$$(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)-3m} \frac{g(\ell)}{\ell-\sigma_a+m}\,d\ell.$$

As $m \to 0$, the upper bound tends to $\sigma_a + (1-\rho_a^0)W(I)$, and the integrand tends to $g(\ell)/(\ell-\sigma_a)$ while staying below $\sup g/\delta$. This lower bound therefore tends to

$$(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)} \frac{g(\ell)}{\ell-\sigma_a}\,d\ell,$$

which grows without bound as δ falls, since $g(\sigma_a) > 0$.

The choice of δ, then of m. Take δ such that this limit exceeds $4W(I)\sup g$, then m small enough: the lower bound of the gain divided by m exceeds the loss divided by m, and the profit strictly rises. Applied to the optimal fixed share, with $\sigma_a^P > 0$ by assumption, this rules out $m^* = 0$. □

Proof of Proposition 19

We show first point (i). The seed investor's profit depends on the protections only through the share they leave it. The founders' stake is written before the seed round and comes first, like the accelerator's protection: by the rule of Section 2, with a fixed share ρ_a the seed investor's share is at most $1-\rho_e-\rho_a$, and with an MFN clause the SAFEs signed after the protections share $1-\rho_e-\rho_a^0$. The seed investor's program is therefore the program of the corresponding proposition with ρ_a replaced by $\rho_e+\rho_a$, or ρ_a^0 by $\rho_e+\rho_a^0$. Without protection, it is the program of Propositions 4 and 5 with $\rho_a = \rho_e$: the seed investor invests its need, takes $1-\rho_e$, finances the shock if and only if $\ell - \sigma_a \le (1-\rho_e)W(I)$, and the cap $(\ell-\sigma_a)/(1-\rho_e) = \sigma_s^*/(1-\rho_e)$ gives it this share in every state above I. With $\sigma_a = 0$ and no protection, the covered projects are the shocks $\ell \le (1-\rho_e)W(I)$, at the threshold I, with surplus $W(I)-\ell$, and the other projects are abandoned with surplus 0, so that

$$TS = \int_0^{(1-\rho_e)W(I)} \big(W(I)-\ell\big)\,g(\ell)\,d\ell = TS^* - \int_{(1-\rho_e)W(I)}^{W(I)} \big(W(I)-\ell\big)\,g(\ell)\,d\ell,$$

where the last integral is positive since $g > 0$ and $\rho_e > 0$.

We show next what the accelerator keeps on a project the seed investor finances, with amount σ_s^*, threshold x^* and profit Π_s^*. In every state where the project is fully funded, the founders receive the share ρ_e of the equity, and the accelerator and the SAFEs share the rest, whose expected value is $(1-\rho_e)W(x^*)$. The seed investor receives $\Pi_s^* + \sigma_s^*$ in expectation, so that the accelerator keeps

$$(1-\rho_e)W(x^*) - \sigma_s^* - \Pi_s^*.$$

We show next point (ii). With a fixed share ρ_a, the seed investor invests its need and takes $1-\rho_e-\rho_a$ by point (i), so that $\Pi_s^* = (1-\rho_e-\rho_a)W(I) - (\ell-\sigma_a)$, and the accelerator keeps $\rho_a W(I)$ on the rounds $\ell - \sigma_a \le (1-\rho_e-\rho_a)W(I)$. Its profit is

$$(1-\rho_e)\int_0^{\sigma_a} W(I+\ell-\sigma_a)\,g(\ell)\,d\ell - \sigma_a + \rho_a W(I)\Big[G\big(\sigma_a+(1-\rho_e-\rho_a)W(I)\big) - G(\sigma_a)\Big],$$

whose derivative in ρ_a at $\rho_a = 0$ is $W(I)[G(\sigma_a + (1-\rho_e)W(I)) - G(\sigma_a)]$, positive since $g > 0$. At the optimal contribution without protection, a small fixed share therefore raises the profit strictly, and the optimal fixed share (σ_a^P, ρ_a^P) earns strictly more than without protection. It follows that $\rho_a^P > 0$, and that $\rho_a^P < 1 - \rho_e$, since with $\rho_a = 1 - \rho_e$ the seed investor finances no shock and the accelerator earns what it earns without protection.

We show next point (iii). In the state $\pi > x$, the SAFE claims the value $\max(\sigma_a(\pi - x)/V_a,\ \sigma_a/\lambda_a)$, and takes at most $(1-\rho_e)(\pi - x)$. If $V_a < \sigma_a/(1-\rho_e)$, the claimed share σ_a/V_a exceeds $1-\rho_e$ in the states $\pi - x \geq V_a/\lambda_a$, and the claimed value σ_a/λ_a exceeds $(1-\rho_e)V_a/\lambda_a > (1-\rho_e)(\pi - x)$ in the others, so that the SAFE takes $1-\rho_e$ in every state and the seed investor finances no shock. If $V_a \geq \sigma_a/(1-\rho_e)$, then $\sigma_a/(\lambda_a(1-\rho_e)) \leq V_a/\lambda_a$, and the three zones of the proof of Proposition 7, with these two thresholds, show that the SAFE leaves

$$(1-\rho_e)\Big(\pi - x - \frac{\sigma_a}{\lambda_a(1-\rho_e)}\Big)^+ - \frac{\sigma_a}{V_a}\Big(\pi - x - \frac{V_a}{\lambda_a}\Big)^+,$$

which is zero below the first threshold, $(1-\rho_e)(\pi - x) - \sigma_a/\lambda_a$ between the two, and $(1 - \rho_e - \sigma_a/V_a)(\pi - x)$ above the second, and whose expectation is $(1-\rho_e)W(x + \sigma_a/(\lambda_a(1-\rho_e))) - (\sigma_a/V_a)W(x + V_a/\lambda_a)$. This value minus σ_s bounds the seed investor's profit, and has derivative

$$(1-\rho_e)\Big(1 - F\Big(x + \frac{\sigma_a}{\lambda_a(1-\rho_e)}\Big)\Big) - \frac{\sigma_a}{V_a}\Big(1 - F\Big(x + \frac{V_a}{\lambda_a}\Big)\Big) - 1 < 0$$

in σ_s, so that the bound is maximal at the need, where a priced SAFE of share $1-\rho_e$, or an unpriced SAFE of cap $(\ell - \sigma_a)/(1-\rho_e)$, which claims at least $1-\rho_e$ in every state, attains it. The seed investor finances the shock if and only if this value at $x = I$ covers $\ell - \sigma_a$, and the accelerator keeps $(1-\rho_e)W(I)$ minus this value. Its profit is therefore the profit of point (ii) with $\rho_a W(I)$ replaced by the value kept.

The fixed shares $\rho_a \in [0, 1-\rho_e]$ allow every value kept in $[0, (1-\rho_e)W(I)]$, which contains the values the SAFE allows, so that no SAFE earns more than the optimal fixed share. With a contribution $\sigma_a > 0$, the SAFE leaves at most $(1-\rho_e)W(I + \sigma_a/(\lambda_a(1-\rho_e))) \leq (1-\rho_e)W(I + \sigma_a/(1-\rho_e))$ at $x = I$, since W decreases and $\lambda_a \leq 1$, so that the value kept is at least $(1-\rho_e)[W(I) - W(I + \sigma_a/(1-\rho_e))]$. This value kept tends to its lower bound with $\lambda_a = 1$ as $V_a \to +\infty$, equals $(1-\rho_e)W(I)$ at $V_a = \sigma_a/(1-\rho_e)$, and is continuous in (V_a, λ_a) on the connected set $V_a \geq \sigma_a/(1-\rho_e)$, $\lambda_a \in (0, 1]$, so that it takes every value in $\big((1-\rho_e)[W(I) - W(I + \sigma_a/(1-\rho_e))],\ (1-\rho_e)W(I)\big]$. If $\sigma_a^P > 0$ and $\rho_a^P W(I)$ exceeds the lower bound at σ_a^P, a SAFE with contribution σ_a^P keeps $\rho_a^P W(I)$ and earns the profit of the optimal fixed share. If $\sigma_a^P = 0$, then $\rho_a^P > 0$ by point (ii), the lower bound tends to 0 with σ_a, so that for σ_a small enough a SAFE with contribution σ_a keeps $\rho_a^P W(I)$, and its profit tends to the profit of the fixed share, since $(1-\rho_e)\int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a$ tends to 0, as in the proof of Proposition 7, and G is continuous.

We show last point (iv). With an MFN clause, the SAFEs signed after the protections share at most $1 - \rho_e - \rho_a^0$, so that the accelerator holds at least the share ρ_a^0 of the

equity, and keeps at least $\rho_a^0 W(x^*) \geq \rho_a^0 W(I)$ on every financed project, which is what it keeps with $m = 0$. On a round where the seed investor invests its need and takes its ceiling $(1-\rho_e-\rho_a^0)(\ell-\sigma_a)/(\ell-\sigma_a+m)$, its profit is this ceiling times $W(I)$, minus $\ell - \sigma_a$, and the accelerator keeps

$$(1-\rho_e)W(I) - (\ell-\sigma_a) - \Pi_s^* = \rho_a^0 W(I) + \frac{(1-\rho_e-\rho_a^0)\, m\, W(I)}{\ell-\sigma_a+m}.$$

The proofs of Propositions 9 (i) and 18 then hold with $1-\rho_a^0$ replaced by $1-\rho_e-\rho_a^0$, where $\rho_a^0 = \rho_a^P$, for the optimal fixed share, lies in $(0, 1-\rho_e)$ by point (ii), and two of their steps use ρ_a^0 otherwise. In the proof of Proposition 9, large excess amounts lose because the integrand $1-(1-\rho_a^0)(1-F)$ is at least ρ_a^0; with the founders' stake, the integrand $1-(1-\rho_e-\rho_a^0)(1-F)$ is at least $\rho_e+\rho_a^0 > 0$. In the proof of Proposition 18, the accelerator keeps $(1-\rho_e)W(x^p) - \sigma_s^p - \Pi_s^p$, and Π_s^p is at most the ceiling times $W(x^p)$, minus σ_s^p, so that the accelerator keeps at least $1-\rho_e$ minus the ceiling, times $W(x^p) \geq W(I) + (1-F(I))(\sigma_s^p - (\ell-\sigma_a))$. The factor $1-\rho_e$ minus the ceiling is at least $\rho_a^0 > 0$, and the derivative argument of that proof shows that this lower bound increases with σ_s^p for small m, so that it is smallest at the need, where it equals the value above. In both proofs, the seed investor's zone ends at $\sigma_a+(1-\rho_e-\rho_a^0)W(I)-m$ instead of $\sigma_a + (1-\rho_e-\rho_a^0)W(I)$, the gain divided by m tends to $(1-\rho_e-\rho_a^0)W(I)$ times the integral of $g(\ell)/(\ell-\sigma_a)$ over the zone, which diverges as the zone extends to σ_a since $g(\sigma_a) > 0$, and the loss on the rounds lost, which lie in bands of widths proportional to m, stays bounded once divided by m. □